

Terraform on Google Cloud Platform (GCP): Complete Beginner to Advanced Guide

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From First Principles to Production-Grade Enterprise Architecture

*A practical, hands-on guide for Cloud Engineers, DevOps Engineers, Platform Engineers, and
Solution Architects*

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About This Guide

This guide takes you from zero Terraform knowledge to designing and operating production-grade Google Cloud Platform (GCP) infrastructure with Terraform. It is written for five audiences at once — complete beginners, Cloud Engineers, DevOps Engineers, Platform Engineers, and Solution Architects — so every concept is first explained simply, then reinforced with a real, runnable GCP example.

How to use this guide:

TIP

If you are brand new to Terraform, read Chapters 1-9 in order before jumping to advanced topics. If you already know Terraform basics but are new to GCP, start at Chapter 4. If you are preparing for an interview, jump straight to Chapter 23.

Every chapter follows the same pattern: a plain-language explanation, an architecture diagram where useful, a complete and runnable Terraform code example, a line-by-line explanation of that code, and — where applicable — a troubleshooting and best-practices section.

Code examples use illustrative values such as project ID `my-gcp-project-123456`, region `us-central1`, and zone `us-central1-a`. Replace these with your own values before running any command.

Chapter 1: Introduction

Infrastructure as Code (IaC) means defining servers, networks, databases, and other cloud resources in plain text files instead of clicking through a web console by hand. Where traditional (“ClickOps”) management relies on an engineer manually configuring resources through a UI, one setting at a time, IaC captures that same configuration as code that can be reviewed, versioned, tested, and re-run reliably, over and over, with the same result.

This distinction matters more as systems grow. A single virtual machine is easy to manage by hand. A platform with dozens of services, spread across multiple environments and regions, is not. IaC is what makes that scale manageable, and Terraform is the tool this book uses to get there on Google Cloud Platform (GCP).

Throughout this book, “infrastructure” covers everything from compute instances and networks to storage buckets, IAM permissions, and managed services like Cloud SQL. Almost anything you can configure by hand in the GCP Console has a corresponding Terraform resource, and by the end of this book you will be comfortable defining all of it as code rather than clicking through menus.

Why Automate Infrastructure

Manual infrastructure management works fine for a weekend project or a single test server, but it breaks down quickly as systems grow. Automating provisioning with code removes repetitive manual steps and makes environments predictable, so the same configuration produces the same result every time, whether it is applied by one engineer or by a whole team. Automation also means infrastructure changes can be tested, reviewed, and rolled back the same way application code is, instead of relying on tribal knowledge and careful memory.

As organizations move faster and adopt multi-region or multi-cloud strategies, the gap between “click a few buttons” and “run a script that provisions everything” becomes the difference between hours and days of work. Teams that automate early tend to scale their infrastructure headcount much more slowly than their infrastructure footprint, because the same handful of engineers can safely manage far more once the manual toil is removed.

Challenges of Manual Management

- **Configuration drift** - manual changes made directly in the console slowly diverge from what was originally documented, until nobody is sure what is actually running or why.
- **Human error** - repetitive point-and-click steps are easy to get wrong, especially under time pressure or during an incident, and small mistakes can cause outages.
- **No audit trail** - there is no reliable record of who changed what, when, or why, which makes debugging and compliance much harder.
- **Slow scaling** - replicating an environment by hand across regions or projects takes hours or days instead of minutes, and mistakes compound with every repetition.

Benefits of Infrastructure as Code

- **Repeatability** - the same code produces identical infrastructure every time it runs, removing the “it worked when I clicked it” problem entirely.
- **Version control** - configurations live in Git alongside application code, so every change has history, a clear author, and can be rolled back if something goes wrong.
- **Review and collaboration** - changes go through pull requests just like code, letting teammates catch mistakes and suggest improvements before they ever reach production.
- **Self-documentation** - the code itself describes exactly what infrastructure exists and how it is configured, removing the need for separate documentation that inevitably goes stale.

Together, these benefits turn infrastructure management from a manual, error-prone chore into an engineering discipline with the same rigor as software development. Testing, code review,

continuous integration, and staged rollouts, all standard practice for application code, become available for infrastructure changes too.

Why Terraform

Terraform, built by HashiCorp, is the most widely adopted IaC tool available today, and for good reason: it is declarative. Instead of writing a script that describes each step to take, you describe the desired end state of your infrastructure, and Terraform's engine figures out the sequence of API calls needed to get there. Several other qualities reinforce this as the right default choice for most teams:

- **Provider ecosystem** - thousands of providers let Terraform manage GCP, AWS, Azure, Kubernetes, and countless SaaS platforms through one consistent workflow and language.
- **Plan and apply safety** - terraform plan shows exactly what will be created, changed, or destroyed before anything is touched, turning risky changes into reviewable ones.
- **State tracking** - Terraform keeps a state file mapping your code to real-world resources, which enables accurate incremental updates and clean, complete teardown.
- **Open source with a large community** - extensive documentation, modules, and community support make it easier to solve problems you will inevitably run into.
- **Cloud-agnostic skills** - the concepts and workflow you learn managing GCP with Terraform transfer directly to AWS, Azure, or any other supported platform, protecting your investment in learning the tool.

Terraform vs. Other Options

GCP offers its own native tool, Deployment Manager, and AWS users are often familiar with CloudFormation. The table below summarizes how Terraform compares.

Aspect	Terraform	CloudFormation	Deployment Manager
Multi-cloud	Yes (AWS, GCP, Azure, etc.)	No (AWS only)	No (GCP only)
Language	HCL (declarative)	JSON/YAML	YAML/Jinja2/Python
State	Explicit state file	Managed by AWS	Managed by GCP
Maturity/notes	Large community, mature ecosystem	Deep native AWS integration	Deprecated by Google in favor of Terraform

Because Google itself now recommends Terraform over Deployment Manager for most new projects, and because most teams run more than one cloud or expect to eventually, Terraform is the safer long-term investment for anyone learning IaC today. CloudFormation remains a solid choice for teams committed entirely to AWS, but it offers no path to managing GCP or Azure resources with the same tool and syntax.

Plugin Architecture

Terraform's core is intentionally lightweight; nearly all cloud-specific logic lives in separate provider plugins that Core calls out to.

The CLI parses your configuration files and hands the work to Core, which builds a dependency graph of every resource and determines the correct order of operations. Core then calls the relevant provider plugin, which translates Terraform's generic "create, update, delete" resource actions into the specific API calls a given cloud understands. That plugin talks to the cloud provider's API, which in turn creates, updates, or deletes the actual resources running in your account.

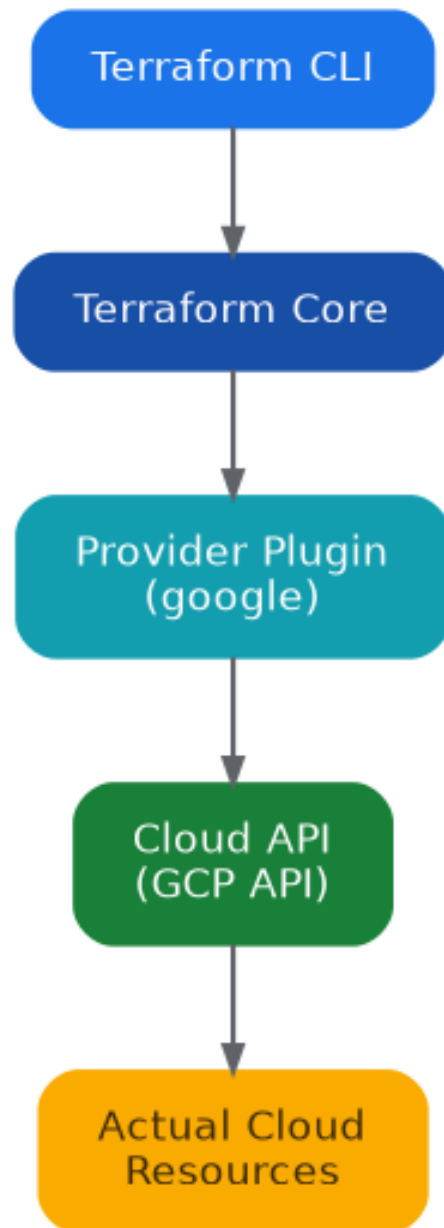


Figure 1: Terraform's layered plugin architecture: the CLI hands work to Core, which delegates to a provider plugin, which calls the cloud API to manage real resources.

TIP

You can combine multiple providers in a single configuration, for example `google` and `google-beta`, to mix stable resources with newer preview features from the same underlying cloud.

This layered design is also why Terraform can support new GCP features quickly: HashiCorp and Google collaborate on the `google` and `google-beta` provider plugins, releasing updates independently of Terraform Core itself. You rarely need to upgrade the whole Terraform binary just to get access to a newly released resource type.

Key Takeaways

- IaC replaces manual, error-prone console work with reviewable, version-controlled code that can be tested and rolled back.
- Manual management suffers from drift, human error, poor auditability, and slow scaling; IaC directly addresses all four problems.
- Terraform's declarative model, large provider ecosystem, and plan/apply workflow make it the leading multi-cloud IaC tool, and Google now recommends it over Deployment Manager.
- Terraform's plugin architecture keeps its core small and stable while providers handle the details of talking to each cloud's specific API.

Chapter 2: Terraform Fundamentals

Every Terraform project follows the same basic workflow: you **write** configuration files describing the infrastructure you want, run **init** to set up the working directory and download the required providers, run **plan** to preview exactly what will change, run **apply** to make those changes real against your cloud account, and eventually run **destroy** to tear the infrastructure down cleanly when it is no longer needed. This write-init-plan-apply-destroy cycle is the foundation for everything else covered in this book, and you will repeat it constantly, often many times a day, as you build out real infrastructure.

Because plan and apply are separate steps, Terraform gives you a chance to review every change before it happens, rather than committing to it blindly. This is one of the biggest practical advantages IaC has over manual console changes, and it is the reason many teams require a reviewed plan output as part of every pull request that touches infrastructure.

Terraform configurations are written in HashiCorp Configuration Language (HCL), a declarative language designed to be readable by humans while still being structured enough for machines to parse reliably. Files use the `.tf` extension, and Terraform automatically loads every `.tf` file in the current directory as part of one combined configuration, regardless of how you split content across files.

Lifecycle Commands

Command	What It Does
<code>terraform init</code>	Initializes the working directory, downloads required providers and modules
<code>terraform validate</code>	Checks configuration syntax and internal consistency without touching real resources
<code>terraform plan</code>	Shows an execution plan: what will be created, changed, or destroyed
<code>terraform apply</code>	Applies the plan, creating or updating real infrastructure
<code>terraform destroy</code>	Removes all resources currently managed by the configuration

You will run `init` once per project (and again whenever you add a new provider or module), `validate` and `plan` frequently while iterating, `apply` whenever you are ready to commit a change, and `destroy` mainly for temporary or throwaway environments. In day-to-day use, most engineers run `plan` many times for every single `apply`, using it as a fast feedback loop while writing configuration.

Core Components

Providers are plugins that let Terraform talk to a specific platform, such as GCP, AWS, or Kubernetes. You declare a provider block once per configuration, and it handles authentication and all the underlying API calls for every resource of that type, so individual resource blocks stay simple.

Resources are the actual infrastructure objects you want Terraform to manage, such as a virtual machine, a storage bucket, or a firewall rule. Each resource block maps to exactly one real-world object, identified by a resource type and a local name that you choose.

Data sources let you read information about existing infrastructure that Terraform did not create, such as looking up an existing VPC network or the latest available OS image. They are strictly read-only and never modify anything, which makes them safe to use liberally.

Variables make configurations reusable by parameterizing values like project ID, region, or machine type instead of hardcoding them directly into resource blocks. This lets the same configuration be reused across environments simply by changing input values. Variables get a full dedicated chapter later in this book.

Outputs expose values from your configuration after it runs, such as a generated IP address or a resource's unique ID, so they can be viewed on the command line or consumed by other configurations. Outputs are also covered in depth in a later chapter.

Modules are reusable, self-contained bundles of Terraform configuration that can be called with different input values, similar to functions in a general-purpose programming language. Modules let teams standardize how common resources, like a VPC or a Kubernetes cluster, get built. A dedicated chapter later explains how to write and consume them.

State is a file Terraform uses to map your configuration to the real resources it manages, tracking metadata about each resource and enabling accurate diffs on every subsequent plan. State has enough nuance around locking, remote backends, and safe migration that it warrants its own full chapter later in this book.

Two more concepts round out the fundamentals you will use every day. **Resource dependencies** are usually inferred automatically: if one resource block references an attribute of another, Terraform knows to create the referenced resource first. **The dependency graph** built from those relationships is what allows Terraform to safely parallelize resource creation wherever there is no ordering requirement, which is part of why large configurations still apply relatively quickly.

Basic GCP Flow

At its simplest, working with GCP through Terraform looks like this:

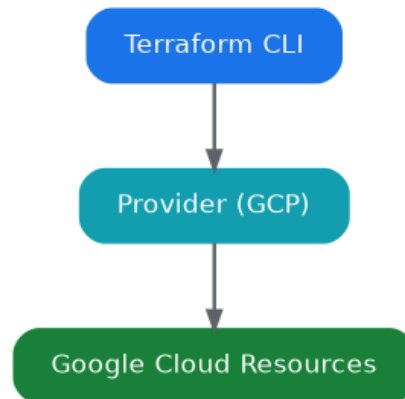


Figure 2: The basic GCP flow: Terraform CLI drives the google provider, which manages Google Cloud resources directly.

The CLI reads your `.tf` files, hands the described work to the configured google provider, and that provider creates, updates, or deletes the corresponding resources inside your Google Cloud project. Every example in this book builds on this same basic shape, just with more resources and more structure layered on top, whether that means adding variables, splitting configuration into modules, or storing state remotely for team collaboration.

Example Configuration

```
terraform {
  required_providers {
    google = {
      source = "hashicorp/google"
      version = "~> 5.0"
    }
  }
}

provider "google" {
  project = "my-gcp-project-123456"
  region = "us-central1"
}

resource "google_storage_bucket" "example" {
  name     = "my-gcp-project-123456-demo-bucket"
  location = "US"
}
```

- The terraform block pins the required provider and constrains it to a compatible version range.
- The provider "google" block sets the default project and region used by every resource declared below it.
- resource "google_storage_bucket" "example" declares one storage bucket; "example" is its local name inside Terraform, not the actual bucket name in GCP.
- name and location are resource-specific arguments that configure the real bucket that gets created.
- Together, these three blocks are a complete, runnable Terraform configuration on their own.

Running terraform init followed by terraform plan against this file shows exactly one resource scheduled for creation; running terraform apply then creates that bucket for real in your GCP project.

BEST PRACTICE

Always run terraform plan and read its output carefully before running apply, especially in shared or production environments, since plan is your last chance to catch an unintended change.

Once applied, running terraform plan again with no further changes to the configuration should report that no changes are needed. If it instead reports differences, something outside Terraform, such as a manual console edit, has caused configuration drift, and Terraform will offer to correct it back to match your code on the next apply.

Key Takeaways

- The core Terraform workflow is write, init, plan, apply, destroy - memorize this cycle, since you will use it constantly.
- init, validate, plan, apply, and destroy are the five lifecycle commands you will run every day.
- Providers, resources, and data sources are the building blocks used in every configuration; variables, outputs, modules, and state each get their own dedicated chapter later.
- Terraform's entire job is to translate your configuration, through a provider, into real API calls against GCP, and state is what lets it do that reliably over time.

Chapter 3: Installing Terraform

Terraform installs the same basic way on every major operating system: download a single binary and make sure it is on your PATH so it can be run from any directory. This chapter walks through the fastest, most reliable method for Windows, Linux, and macOS, and then shows how to verify the install actually worked before moving on to the rest of the book.

Unlike many developer tools, Terraform has no separate runtime or dependency to manage; the compiled binary is entirely self-contained. That simplicity is part of why the installation steps below are so short, and why upgrading later is usually just a matter of replacing that one file, or letting your package manager do it for you.

Windows

The quickest method uses the Chocolatey package manager, which handles downloading, extracting, and adding Terraform to your PATH automatically:

```
choco install terraform
```

If you would rather not use a package manager, you can instead download the Windows zip archive from the official HashiCorp Terraform downloads page, extract the single `terraform.exe` file into a folder such as `C:\terraform`, and then add that folder to your system PATH manually through System Properties so it is available from any command prompt. Either approach ends with the same result: a `terraform` command that works from PowerShell or Command Prompt in any directory.

Linux

On Debian and Ubuntu-based systems, the recommended approach is to add HashiCorp's official apt repository and install Terraform from it, which also makes future upgrades a simple `apt-get` upgrade:

```
sudo apt-get update && sudo apt-get install -y gnupg software-properties-common
↪ curl
curl -fsSL https://apt.releases.hashicorp.com/gpg | sudo gpg --dearmor -o
↪ /usr/share/keyrings/hashicorp-archive-keyring.gpg
echo "deb [signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg]
↪ https://apt.releases.hashicorp.com $(lsb_release -cs) main" | sudo tee
↪ /etc/apt/sources.list.d/hashicorp.list
sudo apt-get update && sudo apt-get install -y terraform
```

This sequence installs the prerequisite tools, imports HashiCorp's signing key so apt can verify package authenticity, registers the HashiCorp repository for your specific distribution release, and finally installs Terraform through the normal apt package manager, exactly like any other system package. On Fedora, CentOS, or other RHEL-based distributions, HashiCorp publishes an equivalent yum repository, and the same overall pattern applies using `dnf` or `yum` instead of `apt-get`.

macOS

On macOS, Homebrew is by far the simplest approach, and also makes upgrades trivial later:

```
brew tap hashicorp/tap
brew install hashicorp/tap/terraform
```

The `tap` command adds HashiCorp's own Homebrew repository, and the `install` command then pulls Terraform specifically from that tap rather than the default Homebrew formulas, ensuring you

get an officially maintained build. Homebrew also takes care of PATH configuration automatically, so no manual setup is required afterward.

Verifying the Installation

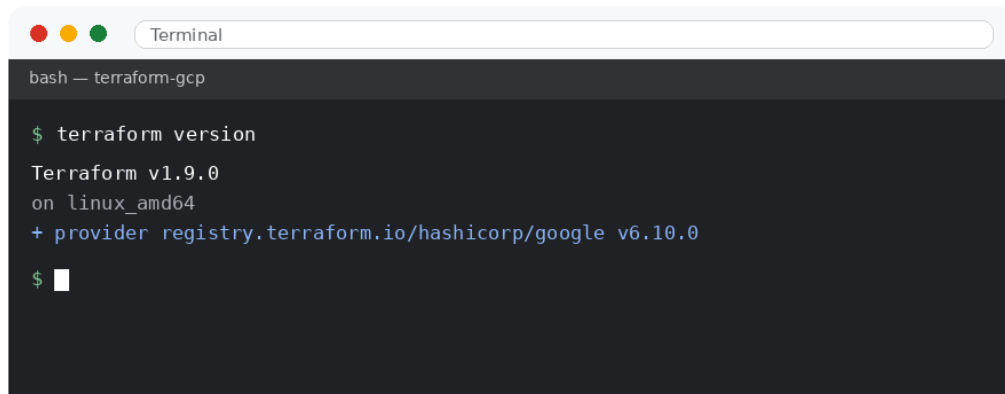
Regardless of which platform you installed on, confirm the install worked with:

```
terraform version
```

Expected output looks similar to:

```
Terraform v1.9.0  
on linux_amd64
```

Seeing a version number here confirms Terraform is installed correctly and is available on your PATH, meaning it can be invoked from any directory on your system, which is exactly what you need for the hands-on exercises in the rest of this book. If the command also warns that a newer version is available, that is informational only; it is safe to continue with the version you have installed unless a specific feature in a later chapter requires upgrading.

A screenshot of a macOS Terminal window titled "Terminal". The prompt is "bash — terraform-gcp". The user has entered the command "\$ terraform version". The output is "Terraform v1.9.0", "on linux_amd64", and "+ provider registry.terraform.io/hashicorp/google v6.10.0". The prompt "\$" is followed by a cursor.

```
bash — terraform-gcp  
  
$ terraform version  
Terraform v1.9.0  
on linux_amd64  
+ provider registry.terraform.io/hashicorp/google v6.10.0  
  
$
```

Figure 3: Illustrative terminal output confirming a working Terraform install and the loaded google provider version.

Choosing an Install Method

There is no single “correct” way to install Terraform, and the right choice mostly depends on how you already manage software on your machine. If you already use Chocolatey, Homebrew, or apt for other developer tools, installing Terraform the same way keeps upgrades and uninstalls consistent with everything else on your system. If you are working in a locked-down environment where you cannot install package managers, the manual zip download works everywhere and requires no special permissions beyond writing to a folder and editing your PATH.

Whichever method you choose on your personal machine, it is worth knowing that most CI/CD systems and Docker-based build pipelines install Terraform differently, typically by pulling an official HashiCorp Docker image or downloading the binary directly in a pipeline step. Chapters later in this book that cover automation will show that approach specifically, so do not worry about it yet.

Troubleshooting

If you see an error like `terraform: command not found` (or an equivalent error on Windows), the binary itself was likely installed correctly but is not on your system’s PATH. Confirm the folder containing `terraform` (or `terraform.exe` on Windows) is actually listed in your PATH environment

variable, then open a brand new terminal session, since existing terminal windows will not pick up PATH changes automatically. On Linux or macOS, you can also confirm the binary exists at all with `which terraform`; on Windows, check under System Properties > Environment Variables that the correct folder was added.

NOTE

If you need to work across multiple Terraform versions on the same machine, which is common when maintaining several projects that were built against different releases, use a version manager like `tfenv` or `tfswitch` instead of reinstalling manually every time you switch projects.

A related but distinct error is seeing an old version number after upgrading. This usually means a stale copy of the binary still exists earlier in your PATH than the newly installed one. Running `which terraform` on Linux/macOS, or `where terraform` on Windows, will list every matching binary in PATH order, making it easy to spot and remove the outdated copy.

Key Takeaways

- Terraform is distributed as a single binary; Chocolatey (Windows), the HashiCorp apt repository (Linux), and Homebrew (macOS) are the fastest and most maintainable install paths.
- `terraform version` is the standard way to confirm a working installation before doing anything else.
- A “command not found” error almost always means a PATH configuration problem, not a broken or corrupted install.
- Use `tfenv` or `tfswitch` if you regularly need to switch between different Terraform versions across projects.

Chapter 4: Google Cloud Prerequisites

Before writing any Terraform code, you need a working Google Cloud environment: an account with billing enabled, a project to hold your resources, a service account Terraform can authenticate as, and the right APIs turned on. None of this is Terraform-specific — it's the same setup any GCP user needs — but getting it right up front saves a lot of confusing “permission denied” and “API not enabled” errors later. This chapter walks through each piece in the minimum detail needed to get moving, and it's the environment every hands-on example in the rest of this book assumes you already have.

Google Cloud Account and Billing

Sign up at cloud.google.com with a Google account. New customers typically receive free trial credits, but any real usage — even a single small virtual machine running for an hour — requires an active billing account linked to a valid payment method. Billing accounts and projects are separate objects: a billing account can pay for many projects, and an organization typically has just one or two billing accounts shared across all its teams.

Create the billing account once in the Console under Billing > Create Account, and link it to every project you create. Without this link, API calls and resource creation will fail with permission or billing errors even when your IAM roles look completely correct — a common source of confusion for newcomers who assume a permissions problem when it's really a missing billing link. If you're working inside an existing company organization, ask whoever manages billing to either link your project to the shared billing account or grant you the `roles/billing.user` role so you can do it yourself.

Creating a Project

Everything in GCP lives inside a project — the container for resources, billing, quotas, and IAM policies. Nothing can be created outside of a project. Create one from the command line with the `gcloud` CLI:

```
gcloud projects create my-gcp-project-123456 \
  --name="Terraform Demo" \
  --organization=123456789012
```

The project ID (`my-gcp-project-123456`) must be globally unique across all of Google Cloud, 6-30 characters long, and cannot be changed after creation — only the display name can be edited later. If you don't belong to an organization, omit `--organization` and the project is created as a standalone, unattached project instead.

You can create the same project through the Console instead, using the “New Project” button on the resource picker at the top of any Console page. This is often easier the first time, since the UI validates the project ID interactively and shows you immediately whether it's available.

IAM Basics

Google Cloud's Identity and Access Management (IAM) system controls who can do what, and it's worth understanding the vocabulary before you start granting access. Three concepts matter most. A **principal** is an identity — a human user, a group, or a service account. A **role** is a named bundle of **permissions**, where each permission maps to a specific API action such as `compute.instances.create` or `storage.buckets.list`. A **binding** connects a role to a principal on a specific resource, whether that's a project, a folder, or the whole organization.

Roles come in three flavors. Basic roles (Owner, Editor, Viewer) predate fine-grained IAM and grant sweeping access across nearly every service — convenient for a quick experiment, but dangerous in any shared or production environment because they violate least privilege. Predefined roles

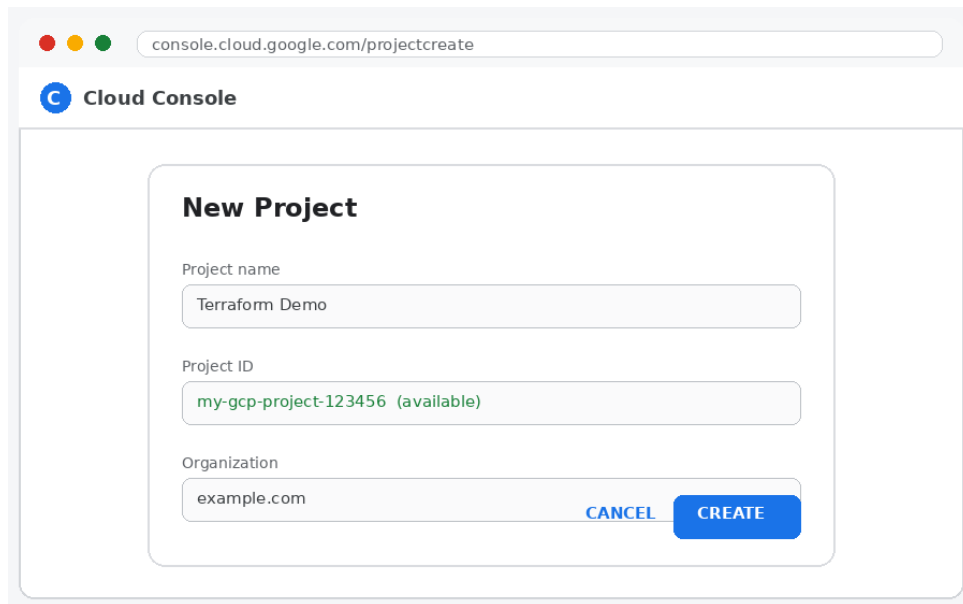


Figure 4: Illustrative “New Project” console dialog, showing the project name, ID availability check, and organization fields.

are scoped to a single service, like `roles/compute.admin` or `roles/storage.objectViewer`, and are curated by Google to match common job functions. Custom roles let you assemble an exact permission set when even the narrowest predefined role is too broad. As a rule, always grant the least privilege necessary for a principal to do its job, and prefer predefined or custom roles over basic ones outside of quick throwaway testing.

Service Accounts for Terraform

Terraform needs its own identity to call GCP APIs on your behalf. You don’t want it running as your personal user account: personal credentials expire, get revoked when someone leaves the team, and make it hard to reason about exactly what automation is allowed to touch. A service account is a non-human identity, created specifically for a program or pipeline, that you grant a defined set of roles.

Create one and bind it to a role like this:

```
gcloud iam service-accounts create terraform-sa \
  --display-name="Terraform Automation"

gcloud projects add-iam-policy-binding my-gcp-project-123456 \
  --member="serviceAccount:terraform-sa@my-gcp-project-
  ↪ 123456.iam.gserviceaccount.com"
  ↪ \
  --role="roles/editor"
```

The first command creates the service account itself; the second grants it a role on the project. In production, replace the broad `roles/editor` shown here with a narrower set of roles that match only the resource types Terraform will actually manage — for example `roles/compute.admin` for VM and network resources, or `roles/storage.admin` if it also manages buckets. This keeps a compromised or misconfigured pipeline from being able to touch anything in the project.

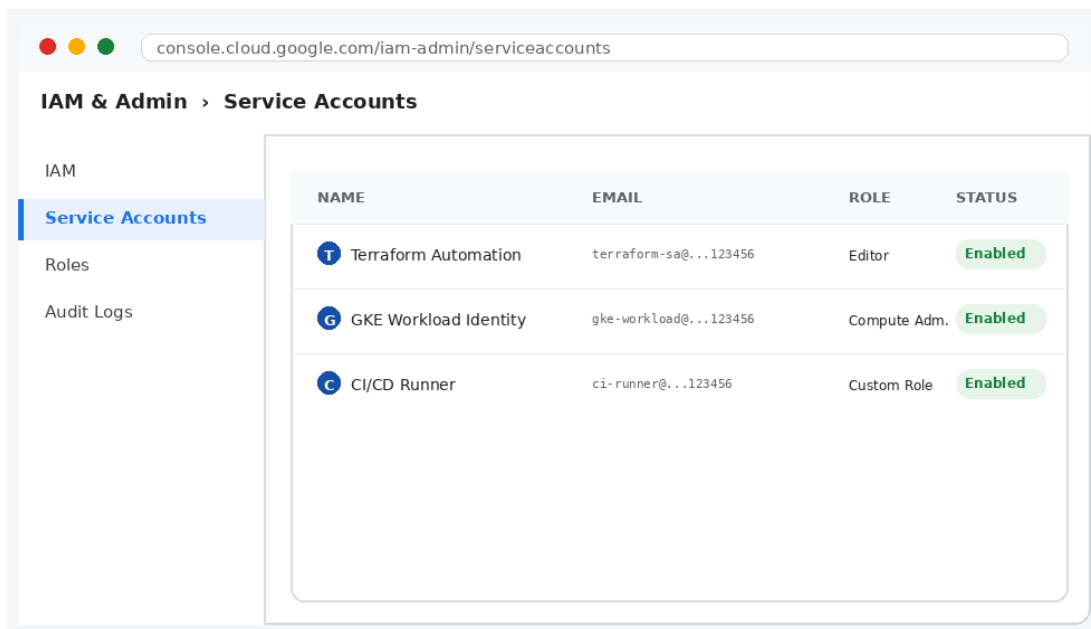


Figure 5: Illustrative IAM & Admin > Service Accounts list, showing the terraform-sa account alongside its assigned role.

Enabling Required APIs

GCP disables most service APIs by default on new projects, and Terraform’s calls will fail with an “API not enabled” error until the relevant service is turned on — even if every IAM role and quota is otherwise correct. Enable the Compute Engine API with:

```
gcloud services enable compute.googleapis.com
```

Depending on what your Terraform configuration manages, you’ll typically also need `iam.googleapis.com` (for managing service accounts and roles), `storage.googleapis.com` (for Cloud Storage buckets), and `cloudresourcemanager.googleapis.com` (for project-level operations like IAM bindings). Enabling every API your configuration touches before your first terraform apply avoids stopping halfway through a large plan with a confusing error.

TIP

Run `gcloud services enable` for every API your configuration touches before your first terraform apply — Terraform will not enable APIs on your behalf, and a mid-apply failure can leave some resources created and others not.

Putting It Together

By the end of this chapter you should have four things ready before opening an editor and writing your first `.tf` file: a project linked to an active billing account, a clear mental model of principals, roles, and bindings, a dedicated terraform-sa service account holding only the roles it needs, and every API your configuration will call already enabled. Skipping any one of these tends to surface as a cryptic error partway through your first terraform apply rather than up front, so it’s worth spending the extra few minutes here rather than debugging blind later. The next chapter picks up exactly where this one leaves off — wiring this service account and project into an actual Terraform provider configuration.

Key Takeaways

- A billing-linked project is the mandatory container for all GCP resources; missing billing links look like permission errors but aren't.
- IAM grants roles — bundles of permissions — to principals; always favor least-privilege pre-defined or custom roles over basic ones.
- Terraform should authenticate as a dedicated service account, never as a personal user identity, so access stays auditable and revocable.
- Every API your configuration calls must be explicitly enabled per project before Terraform can use it.

Chapter 5: Configuring Terraform for GCP

Terraform needs credentials to talk to Google Cloud’s APIs before it can plan or apply anything. Getting authentication right is the first real obstacle most newcomers hit, because GCP supports several methods that suit different situations, and picking the wrong one for your context leads to either a frustrating setup experience or a genuine security risk. This chapter compares the options and then builds the provider configuration you’ll reuse throughout the rest of the book.

Authentication Methods Compared

Method	Best For	Security Level	How It Works
Application Default Credentials (ADC)	Local development	Medium	Uses credentials from <code>gcloud auth application-default login</code> , cached locally on disk
Service Account Key (JSON)	Legacy CI/CD, non-GCP hosts	Low	Static private key file referenced by Terraform; never expires automatically
Workload Identity Federation	CI/CD and workloads running on GCP or federated external systems	High	Exchanges a short-lived external or platform token for GCP credentials; no long-lived secret is stored

For day-to-day development on your own laptop, ADC is the simplest and safest starting point: run `gcloud auth application-default login` once, authenticate through the browser flow, and the `google` provider picks up the resulting credentials automatically with no extra configuration in your Terraform files. Service account key files still work everywhere and are the easiest to understand, which is exactly why they were the default choice for years — but that same portability is what makes them risky once a project has more than one contributor or pipeline. Workload Identity Federation is the modern answer for automation: an external identity provider (GitHub Actions, GitLab CI, or another cloud) proves who it is, GCP issues a short-lived token in exchange, and no static secret ever needs to be generated, stored, or rotated.

The Provider Block

Every GCP Terraform configuration starts with a provider block that tells Terraform which project, region, and credentials to use for every resource that doesn’t override them explicitly:

```
provider "google" {
  project = "my-gcp-project-123456"
  region  = "us-central1"
}
```

- `project` — the GCP project ID that resources are created in by default; individual resource blocks can still set their own `project` argument to override this for multi-project configurations.
- `region` — the default region used for regional resources such as subnets, regional persistent disks, and many managed services; without it, every regional resource block would need its own `region` argument.

- `zone` (optional) — the default zone for zonal resources like Compute Engine instances and zonal disks; commonly set alongside `region` so you don't repeat it on every `google_compute_instance` block.
- `credentials` (optional) — a path to a service account key file, used only when ADC or the GCP metadata server isn't available, such as when running Terraform from a machine outside Google Cloud without federation configured.
- `impersonate_service_account` (optional) — lets a human user or a CI identity temporarily assume a service account's permissions for the duration of a run, without ever holding that service account's key, combining the convenience of a single shared identity with the auditability of per-user authentication.

With just `project` and `region` set, and ADC already configured on your machine, `terraform plan` and `terraform apply` will authenticate automatically for local work — no `credentials` block is required in the common case. This is worth remembering because many older tutorials and examples show a `credentials` argument pointing at a downloaded JSON key, which is no longer the recommended default.

WARNING

Downloaded service account JSON keys are long-lived secrets that sprawl easily across laptops, shell history, CI logs, and version control, and they don't expire on their own unless you rotate or revoke them manually. For CI/CD pipelines, prefer Workload Identity Federation, which issues short-lived tokens on demand and eliminates the key file — and the risk of it leaking — entirely.

Choosing Between the Three in Practice

A simple rule of thumb covers most real projects: use ADC on your own machine while developing and testing configurations interactively, use Workload Identity Federation for any pipeline that runs Terraform automatically (CI/CD, scheduled jobs, GitOps controllers), and reserve service account keys for the narrow case of a system that genuinely cannot support federation — some third-party tools and older on-premises schedulers still fall into this category. If you do end up using a key file, store it in a secrets manager rather than committing it anywhere, and set a calendar reminder to rotate it periodically since GCP does not enforce expiration on your behalf.

Verifying the Configuration

Once the `provider` block is in place, confirm authentication actually works before writing any resources. Running `terraform init` downloads the `google` provider plugin and validates that the block is syntactically correct, but it does not verify credentials. A quick `terraform plan` against a trivial data source, such as `data "google_project" "current" {}`, is a fast way to confirm Terraform can reach the API with the identity you expect — if it fails, the error message will usually tell you directly whether it's a missing API, an authentication problem, or an IAM permission gap. It's worth doing this check once per environment (local, CI, staging) rather than assuming that because it worked on your laptop it will also work in the pipeline, since each environment may resolve credentials through a different one of the three methods described above.

Provider Version Pinning

One detail worth setting alongside the `provider` block itself is the version constraint, declared in a `terraform` block:

```
terraform {
  required_providers {
    google = {
      source = "hashicorp/google"
      version = "~> 5.0"
```

```
}  
}  
}
```

Without a version constraint, `terraform init` downloads whatever the latest provider release happens to be at that moment, which means the exact same configuration can behave differently depending on when and where it's initialized. The `~> 5.0` syntax allows any 5.x release but blocks an accidental jump to a breaking 6.x major version. Combined with the `.terraform.lock.hcl` file discussed in the next chapter, this gives every teammate and every CI run the same provider behavior, which matters more than it might seem the first time a provider upgrade silently changes a resource's default value.

Key Takeaways

- GCP offers three authentication paths — ADC, service account keys, and Workload Identity Federation — trading setup convenience for security differently.
- A minimal provider `"google"` block needs only `project` and `region`; `zone`, `credentials`, and `impersonate_service_account` are optional refinements for specific situations.
- ADC is the right default for local development and requires no key files or extra provider configuration.
- Avoid static service account keys in CI/CD wherever possible — Workload Identity Federation is the safer, modern standard for automated pipelines.

Chapter 6: Terraform Project Structure

A consistent file layout makes Terraform configurations easier to read, review, and hand off — for you six months from now, and for anyone who joins the project later without any context on how it was built. Terraform itself doesn't enforce any particular file structure; it simply loads and merges every `.tf` file in a directory into one configuration. The layout in this chapter is a widely adopted convention, used throughout the rest of this book, that keeps concerns separated so a reviewer can find what they're looking for without reading the whole configuration top to bottom.

The Standard Layout

```
terraform-gcp/  
├─ main.tf  
├─ variables.tf  
├─ outputs.tf  
├─ provider.tf  
└─ terraform.tfvars
```

- `main.tf` — holds the core resource and module blocks that define what actually gets built, such as `google_compute_instance` or `google_storage_bucket` resources; this is usually the file people read first to understand what a configuration does.
- `variables.tf` — declares input variables with a name, type, description, and optional default, without assigning any environment-specific values; think of it as the configuration's parameter list.
- `outputs.tf` — declares values Terraform should print to the terminal or expose to other configurations after a successful apply, such as an instance's external IP address or a generated bucket name.
- `provider.tf` — configures the `google` provider block along with any `terraform { }` block settings, such as the required Terraform and provider versions and the backend used for remote state.
- `terraform.tfvars` — supplies the actual values for the variables declared in `variables.tf`, kept in a separate file so the same `.tf` files can be reused unchanged across different environments by swapping only this one file.

Because Terraform merges every `.tf` file in a directory regardless of its name, this split is purely a human convention rather than a technical requirement — you could put everything in one giant file and Terraform would behave identically. Following the convention anyway pays off quickly: a reviewer looking for outputs doesn't have to scroll through resource definitions, and a teammate changing an environment's settings only needs to touch `terraform.tfvars`.

Scaling Beyond a Single Directory

This flat, five-file layout works well for a single environment managed by one team. As a project grows, two extensions become common. First, teams extract repeated groups of resources into a `modules/` directory — for example `modules/network` or `modules/gke-cluster` — so the same networking or cluster pattern can be reused across multiple projects without copy-pasting resource blocks. Second, teams create one directory per environment, such as `environments/dev`, `environments/staging`, and `environments/prod`, each with its own `terraform.tfvars`, its own state, and its own `provider.tf` pointing at the right project — but all calling the same shared modules. This keeps the difference between environments limited to variable values and instance sizing rather than duplicated resource logic, which in turn keeps dev and production configurations from silently drifting apart over time.

Ignoring Local State and Secrets

Terraform generates local files on every run that should never be committed to version control. State files record every resource's attributes in plaintext, including in some cases sensitive values, and the `.terraform/` directory holds downloaded provider binaries that are large and machine-specific. A `.gitignore` for a Terraform project typically looks like this:

```
.terraform/
*.tfstate
*.tfstate.*
crash.log
*.tfvars
override.tf
override.tf.json
*_override.tf
*_override.tf.json
.terraformrc
terraform.rc
```

Note that `.terraform.lock.hcl` is deliberately left off this list — it should be committed, not ignored, since it pins the exact provider versions and checksums Terraform resolved, ensuring every teammate and every CI run downloads identical provider binaries instead of silently drifting onto a newer version.

Naming and Formatting Conventions

Beyond the file layout itself, a few small conventions keep configurations readable as they grow. Resource names should use `snake_case` and describe purpose rather than type — `google_compute_instance.web_server` reads better than `google_compute_instance.instance1`, since the resource type is already visible in the block header. Run `terraform fmt` before every commit; it rewrites indentation and alignment automatically and removes an entire category of noisy diffs in code review. Run `terraform validate` alongside it to catch syntax and type errors locally, before they surface as a slower failure during `terraform plan` in CI. Together these two commands take seconds to run and catch the large majority of avoidable mistakes before a teammate ever sees the pull request.

A Note on Backend Configuration

The `provider.tf` file is also the conventional home for the backend block, which tells Terraform where to store its state file — locally on disk by default, or remotely in a Google Cloud Storage bucket for anything beyond solo experimentation. Remote state matters as soon as more than one person or one CI job runs Terraform against the same infrastructure, since it provides locking that prevents two applies from racing each other and corrupting state. This book covers remote state and locking in detail in a later chapter; for now, it's enough to know that the backend block lives alongside the provider configuration, not scattered elsewhere in the project.

Key Takeaways

- Terraform merges every `.tf` file in a directory; splitting into `main.tf`, `variables.tf`, `outputs.tf`, and `provider.tf` is a readability convention, not a technical requirement.
- `terraform.tfvars` isolates environment-specific values from reusable resource definitions, letting the same `.tf` files serve multiple environments.
- Larger projects extract shared logic into `modules/` and separate environments under `environments/`, keeping differences limited to variables rather than duplicated code.
- Ignore state files, `.terraform/`, and `.tfvars` in git, but always commit `.terraform.lock.hcl` for reproducible provider versions across machines and CI.

Chapter 7: First Terraform Project

This chapter builds your first real Terraform project on GCP: a custom-mode VPC, one subnet, and a firewall rule allowing SSH and ICMP. It ties together the provider setup and authentication covered in earlier chapters into a single, working, applicable configuration — the same three-resource pattern (network, subnetwork, firewall) you will reuse as the foundation for almost every later chapter in this book. By the end of the chapter you will have created real billable resources in your GCP project, inspected them, and torn them back down cleanly.

Project Layout

Create a directory named `first-project/` and place a single file inside it called `main.tf`. Terraform automatically loads every `.tf` file in the working directory, so file names and file count are a matter of convention rather than a hard requirement — some teams split configuration into `main.tf`, `variables.tf`, and `outputs.tf` even for small projects, while others keep everything together until the file genuinely becomes hard to navigate. For a project this size, one file is easiest to read and easiest to review in a pull request, so that is the approach used here. Later chapters (modules, multi-environment layouts) introduce the split-file convention once there is enough content to justify it.

```
terraform {
  required_providers {
    google = {
      source = "hashicorp/google"
      version = "~> 5.0"
    }
  }
}

provider "google" {
  project = "my-gcp-project-123456"
  region = "us-central1"
}

resource "google_compute_network" "vpc" {
  name = "first-project-vpc"
  auto_create_subnetworks = false
}

resource "google_compute_subnetwork" "subnet" {
  name = "first-project-subnet"
  ip_cidr_range = "10.10.0.0/24"
  region = "us-central1"
  network = google_compute_network.vpc.id
}

resource "google_compute_firewall" "allow_ssh_icmp" {
  name = "allow-ssh-icmp"
  network = google_compute_network.vpc.id

  allow {
    protocol = "tcp"
    ports = ["22"]
  }
}
```

```

allow {
  protocol = "icmp"
}

source_ranges = ["0.0.0.0/0"]
target_tags   = ["ssh-icmp"]
}

```

WARNING

`source_ranges = ["0.0.0.0/0"]` opens SSH to the entire internet. Fine for a learning project; in production restrict it to a bastion IP range or use Identity-Aware Proxy (see Chapter 15).

Explanation

- `auto_create_subnetworks = false` makes this a custom-mode VPC, so you control subnet ranges and regions explicitly instead of relying on GCP's default auto-mode subnets, which pre-allocate a /20 in every region whether you need it or not.
- The subnet references `google_compute_network.vpc.id`, creating an implicit dependency graph — Terraform parses the reference, determines the network must exist first, and orders the create calls accordingly without you writing any explicit ordering logic.
- The firewall rule uses two allow blocks (tcp/22 and icmp) inside one resource rather than two separate `google_compute_firewall` resources, keeping related access rules for the same purpose together and reducing the number of objects to manage.
- `target_tags = ["ssh-icmp"]` means the rule only applies to instances carrying the network tag `ssh-icmp`, not every VM in the VPC — a small habit that prevents a single broad rule from quietly applying to future, unrelated instances.
- All three resources live in one apply, so a single terraform plan shows the complete picture — network, subnet, and firewall — before anything is created, rather than applying them piecemeal and hoping the ordering works out.
- Resource names (`first-project-vpc`, `first-project-subnet`) are GCP-visible identifiers, distinct from the Terraform resource addresses (`google_compute_network.vpc`) used only within the configuration — keep both consistent and descriptive as your project grows.
- No explicit `depends_on` is needed here; explicit dependencies are reserved for cases where one resource affects another without a direct attribute reference, covered in a later chapter.

Running It

Command	Purpose
<code>terraform init</code>	Downloads the Google provider plugin, initializes the backend, and sets up the <code>.terraform</code> working directory
<code>terraform validate</code>	Checks syntax and internal consistency (types, references) without contacting GCP or requiring credentials
<code>terraform plan</code>	Computes and displays what will be created, changed, or destroyed, without touching real infrastructure
<code>terraform apply</code>	Re-runs the plan and, after you confirm, actually creates the resources in GCP

```
terraform init
terraform validate
terraform plan
terraform apply
```

Terraform will perform the following actions:

```
# google_compute_network.vpc will be created
# google_compute_subnetwork.subnet will be created
# google_compute_firewall.allow_ssh_icmp will be created
```

Plan: 3 to add, 0 to change, 0 to destroy.

...

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Read the plan output carefully before typing yes: the leading symbol on each line (+ for create, ~ for update in place, - for destroy) tells you the action Terraform intends to take, and a -/+ pair means a resource will be destroyed and recreated — often because you changed an attribute that can't be updated in place, such as a network's name. Getting comfortable reading this output now will save you from surprises once configurations grow larger and touch resources that are expensive or slow to recreate.

The resource relationships in this project are simple but worth visualizing once, since every later chapter builds on the same shape — a network containing one or more subnets, with firewall rules attached at the network level and scoped down by tags:

After apply finishes, confirm the resources actually exist in GCP rather than trusting the CLI output alone — run `gcloud compute networks list` and `gcloud compute firewall-rules list`, or open the VPC network page in the Cloud Console. This habit catches the rare case where apply reports success but a resource was created with unexpected defaults, and it is good practice to build now before configurations grow too large to eyeball quickly.

Once you have confirmed the three resources exist, run `terraform destroy` to remove all three and stop any associated billing. `destroy` also shows a plan first (this time with - for every resource) and requires the same yes confirmation as `apply`, so it is just as safe to review before committing.

TIP

Run `terraform plan` before every `apply` out of habit, and run `terraform fmt` before committing — it auto-formats indentation and alignment so diffs in code review stay focused on actual changes.

If you plan to keep experimenting rather than destroying immediately, note that re-running `terraform apply` with no configuration changes is safe and produces “No changes” — Terraform is idempotent, so applying the same configuration repeatedly never duplicates resources. This property is what makes the workflow reliable in CI: a pipeline can run `apply` on every merge without needing to track whether infrastructure already exists.

Key Takeaways

- A minimal GCP project needs three resources: network, subnetwork, and firewall — each explicitly linked via `.id` references so Terraform can order their creation correctly.
- Custom-mode VPCs (`auto_create_subnetworks = false`) give you full control over subnet ranges and regions instead of GCP's default auto-mode allocation.
- `init` → `validate` → `plan` → `apply` is the core Terraform workflow you will repeat, largely unchanged, in every chapter that follows.
- Always scope firewall `source_ranges` and `target_tags` as tightly as the use case allows,

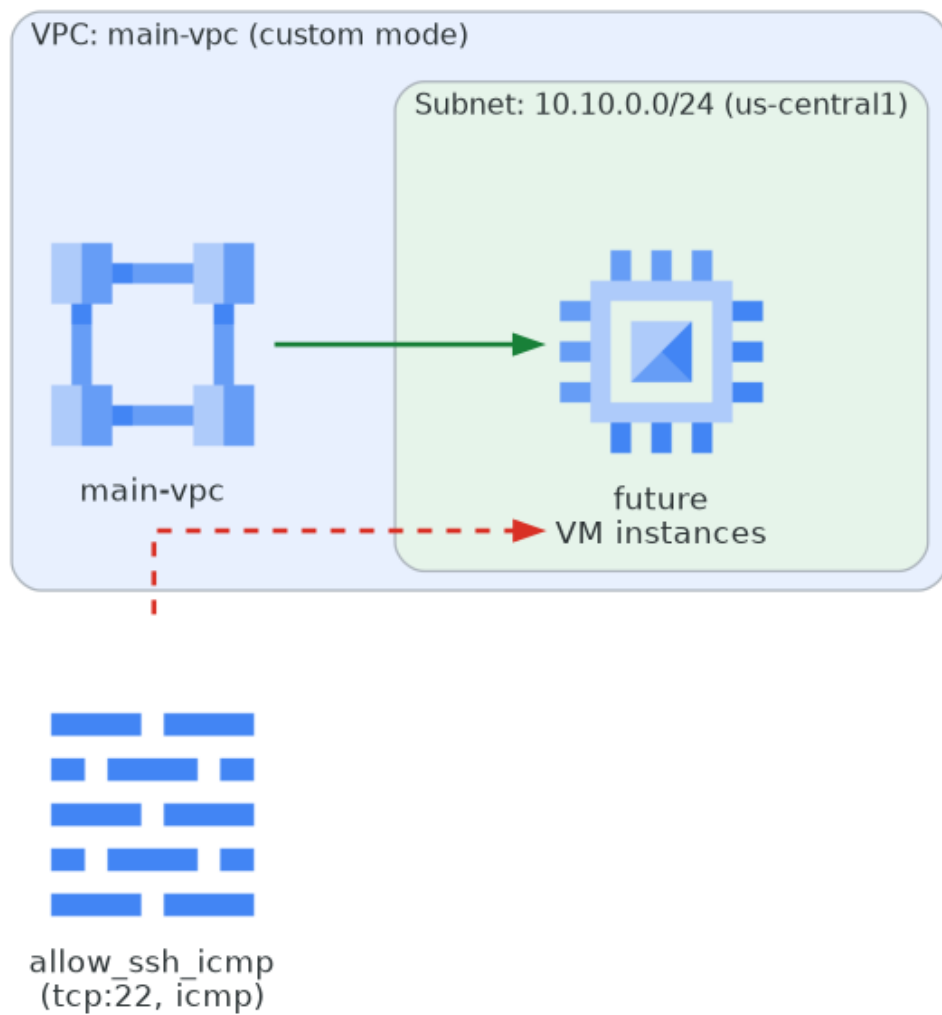


Figure 6: The VPC, subnet, and firewall rule created in this chapter, with the future VM placement they will protect.

and always run `destroy` on throwaway learning projects to avoid ongoing charges.

Troubleshooting tip: if `apply` fails with a “quota exceeded” or “API not enabled” error, run `gcloud services enable compute.googleapis.com` for your project and retry — most first-run failures are a disabled API, not a configuration bug.

Chapter 8: Terraform State Management

Terraform's state file (`terraform.tfstate`) is a JSON record mapping every resource block in your configuration to the real-world object it manages — a GCP instance ID, a VPC self-link, a bucket name, a firewall rule's fingerprint. Without state, Terraform would have no way to know what it already created, so every `plan` and `apply` reads and updates this file to compute the difference between your configuration and reality, then translates that difference into the minimal set of GCP API calls needed to reconcile the two. Losing or corrupting the state file is one of the most common ways teams get into trouble with Terraform, because Terraform then loses track of resources it previously created — it may try to recreate them from scratch, producing duplicate infrastructure or naming conflicts, and the only realistic fix at that point is a careful, manual `terraform import` of every affected resource.

State locking prevents two people (or two CI jobs) from running `apply` against the same state file simultaneously, which could otherwise corrupt the file or produce conflicting, half-applied changes; local state has no locking at all, which is one reason it is unsuitable for anything beyond solo experimentation. State security matters just as much as locking: the state file often contains resource attributes in plain text, including database passwords, TLS private keys, or service account keys generated by providers, even when the corresponding variable was marked sensitive. As a result, state must never be committed to git, must never be pasted into a chat or ticket for debugging, and access to the storage location holding it should be restricted with the same care you would apply to a secrets vault.

Local state (the default `terraform.tfstate` file written into your working directory) also does not scale to teams — two engineers with two local copies of state will inevitably drift out of sync the moment either one runs `apply` without the other's knowledge, and there is no mechanism to reconcile the resulting conflict automatically. Remote state, stored in a shared backend such as Google Cloud Storage, gives every engineer and every CI pipeline the same single source of truth, adds locking so concurrent runs fail safely instead of corrupting data, and removes the need for anyone to carry a personal copy of state on their laptop at all.

GCS Backend

```
terraform {
  backend "gcs" {
    bucket = "my-terraform-state-bucket"
    prefix = "terraform/state"
  }
}
```

Explanation

- The GCS bucket (`my-terraform-state-bucket`) must already exist before you run `terraform init` — Terraform will not create its own backend bucket for you, so provision it once, either manually with `gsutil mb` or `gcloud storage buckets create`, or via a small separate bootstrap configuration that itself uses local state.
- Enable **object versioning** on the bucket (`gsutil versioning set on gs://my-terraform-state-bucket`) so that if a bad `apply` corrupts state, you can retrieve a previous object version rather than losing history entirely.
- GCS backends provide built-in locking via object metadata during `plan` and `apply` — no separate lock service (like DynamoDB for the AWS S3 backend) is required, which is one advantage of using GCS specifically on GCP.
- `prefix` namespaces state within the bucket, letting you store state for multiple projects or environments in one bucket (for example `terraform/state/prod` and `terraform/state/dev`) instead of provisioning a separate bucket per environment.
- Restrict IAM permissions on the bucket to only the humans and service accounts that actually

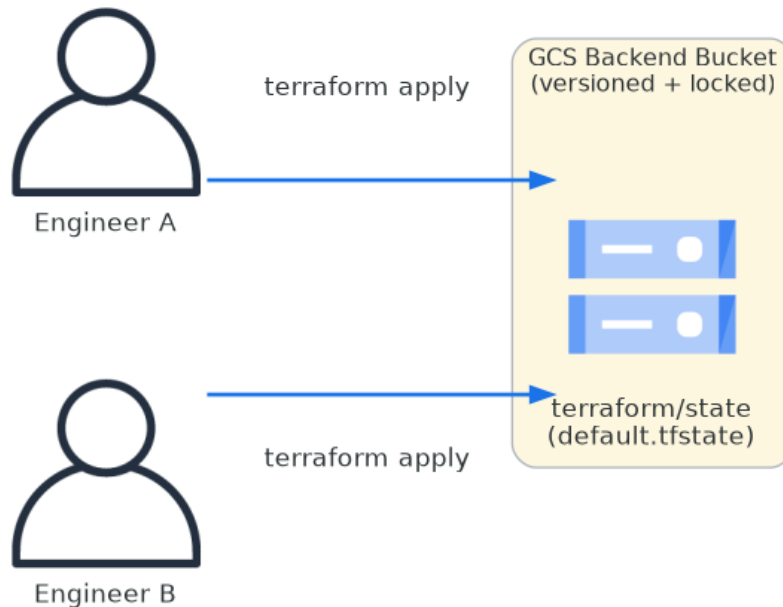


Figure 7: Remote state on GCS: every engineer's `terraform apply` reads and writes the same versioned, locked bucket instead of a personal local copy.

need to run Terraform — `roles/storage.objectAdmin` scoped to that single bucket is usually sufficient, and broader project-wide storage roles should be avoided.

A real state file, once populated, looks roughly like this internally — useful to recognize if you ever need to read one directly for debugging:

```
{
  "version": 4,
  "terraform_version": "1.7.0",
  "resources": [
    {
      "type": "google_compute_network",
      "name": "vpc",
      "instances": [{ "attributes": { "id": "...", "self_link": "..." } }]
    }
  ]
}
```

BEST PRACTICE

Add `*.tfstate*` to `.gitignore` on day one of every project, even if you plan to move to remote state immediately — it is an easy line to forget once deadline pressure sets in, and a state file committed even once remains in git history forever unless the history is rewritten.

If you already have local state and want to switch to this backend, add the block above to your configuration and run:

```
terraform init -migrate-state
```

Terraform detects the new backend configuration, offers to copy the existing local state into the GCS bucket, and asks for confirmation before making any changes — nothing is deleted or overwritten silently.

Terraform supports several backend types beyond local and GCS — Terraform Cloud, an S3-compatible bucket, or Consul among them — but for a GCP-native project, GCS is almost always the simplest choice: no extra service to run, IAM controls access using the same roles you already manage everywhere else in the project, and cost is negligible for a file that is typically a few kilobytes to a few megabytes. If you also want the collaboration features of Terraform Cloud (a web UI, run history, policy checks), it can act as the backend instead while GCP still hosts the actual infrastructure — a decision covered further in Chapter 20 on CI/CD.

Inspecting and Editing State

Use the `terraform state` family of subcommands to inspect or adjust state without hand-editing the raw JSON, which is error-prone and unsupported:

Command	Purpose
<code>terraform state list</code>	Lists every resource address currently tracked in state
<code>terraform state show <addr></code>	Shows the full set of attributes Terraform has recorded for one resource
<code>terraform state rm <addr></code>	Removes a resource from state without destroying it in GCP
<code>terraform state mv <src> <dst></code>	Renames a resource address in state, e.g. after refactoring a module

```
$ terraform state list
google_compute_network.vpc
google_compute_subnetwork.subnet
google_compute_firewall.allow_ssh_icmp
```

`terraform state rm` is useful when you want Terraform to “forget” a resource — for example, before importing it under a different resource name or moving it into a module — without deleting the actual infrastructure running in GCP. Used carelessly, though, it can leave real resources unmanaged and forgotten, so always follow it with `terraform state list` to confirm the remaining inventory matches expectations.

Key Takeaways

- State maps your configuration to real GCP resources; it must be treated as a critical, sensitive artifact, not a disposable log file.
- Never commit `.tfstate` files to git — they can contain secrets in plain text even when the source variable was marked sensitive.
- Remote state (e.g. GCS with versioning enabled) adds team-wide sharing and locking that local state cannot provide.
- Use `terraform state list/show/rm/mv` to inspect or safely adjust tracked resources instead of hand-editing the JSON.

Troubleshooting tip: a “state locked” error that persists after a crashed apply can be cleared with `terraform force-unlock <lock-id>` — but confirm no other apply is genuinely still running first, or you risk two concurrent writes corrupting the file.

Chapter 9: Variables and Outputs

Hardcoding values like project IDs, regions, or CIDR ranges directly into resource blocks makes configurations inflexible, hard to reuse across environments, and risky to change — every edit means touching the resource definitions themselves. This chapter covers Terraform’s four mechanisms for parameterizing configuration: input variables, local variables, environment variables, and output values, extending the VPC project from Chapter 7 as a running example throughout.

Input Variables

Declare variables with a type constraint and optional default in `variables.tf`, then supply real values via `terraform.tfvars`. Terraform automatically loads any file named exactly `terraform.tfvars` (or matching `*.auto.tfvars`) from the working directory with no extra flag required.

```
# variables.tf
variable "project_id" {
  type      = string
  description = "GCP project ID"
}

variable "region" {
  type      = string
  default   = "us-central1"
}

variable "subnet_cidr" {
  type      = string
  default   = "10.10.0.0/24"

  validation {
    condition     = can(cidrhost(var.subnet_cidr, 0))
    error_message = "subnet_cidr must be a valid CIDR block."
  }
}

variable "db_password" {
  type      = string
  sensitive = true
}

# terraform.tfvars
project_id = "my-gcp-project-123456"
subnet_cidr = "10.10.0.0/24"
```

Reference variables elsewhere in the configuration as `var.project_id`. You can also pass values on the command line with `-var="region=us-east1"`, or point Terraform at an alternate file with `-var-file="prod.tfvars"` when managing more than one environment from the same configuration. Marking a variable `sensitive = true` (like `db_password`) redacts it from `plan`/`apply` console output and error messages, though it is still stored in plain text in the state file — sensitivity is a display-time protection, not encryption. The optional validation block on `subnet_cidr` rejects a bad value at `plan` time with a clear error message, rather than letting an invalid CIDR fail deep inside a provider call.

Local Variables

locals define named expressions computed once and reused throughout a configuration — useful for derived values, computed defaults, or reducing repetition across many resources, but unlike `var.*` they are not configurable from outside the configuration; they exist purely to keep the configuration itself readable.

```
locals {
  common_tags = ["managed-by-terraform", "chapter-09"]
  vpc_name     = "${var.project_id}-vpc"
}
```

Reference them the same way as variables, but with the `local.` prefix: `local.common_tags`. A common pattern is combining several variables into one local (as with `vpc_name` above) so every resource that needs the computed name references it once, instead of repeating the same string interpolation in five different resource blocks. Locals also work well for complex types — a `map(string)` of per-environment settings or a `list(object({...}))` describing several subnets is often easier to read as a single `locals` block than as one enormous variable default, and it keeps the externally-configurable surface (variables) separate from internal bookkeeping (locals).

Environment Variables

Terraform reads any environment variable prefixed `TF_VAR_` as the value for the input variable of the same name, which is convenient for CI/CD pipelines that should never write secrets to disk as a `.tfvars` file:

```
export TF_VAR_db_password="s3cr3t-from-ci"
terraform apply
```

This is functionally equivalent to setting `db_password` in a `.tfvars` file, but keeps the secret entirely out of any file that could be accidentally committed, backed up, or left on a shared filesystem. Most CI systems (Cloud Build, GitHub Actions) let you inject `TF_VAR_*` values from their own secret store at pipeline run time, so the secret never touches version control at any point in its lifecycle.

Output Values

Outputs expose resource attributes after `apply` completes, useful for confirming what was created, feeding values into another Terraform configuration, or displaying connection details for humans. Extending Chapter 7's VPC:

```
# outputs.tf
output "vpc_self_link" {
  value      = google_compute_network.vpc.self_link
  description = "Self-link of the created VPC"
}

output "subnet_cidr" {
  value = google_compute_subnetwork.subnet.ip_cidr_range
}

output "db_password" {
  value      = var.db_password
  sensitive = true
}
```

Run `terraform output` after `apply` to view all outputs at once, or `terraform output vpc_self_link` to fetch a single value — handy for scripting, e.g. piping the self-link into a

follow-on `gcloud` command. Sensitive outputs are hidden by default in both forms and require `terraform output -json` (which still marks them, but includes the value) or explicit naming to actually reveal the underlying value.

```
$ terraform output
subnet_cidr    = "10.10.0.0/24"
vpc_self_link  =
↳ "https://www.googleapis.com/compute/v1/projects/my-gcp-project-123456/..."
db_password    = <sensitive>
```

Explanation

- Type constraints (`string`, `number`, `bool`, `list(string)`, `map(string)`, etc.) catch mistakes at plan time rather than deep inside a provider error that is harder to trace back to its source.
- `terraform.tfvars` is auto-loaded with no flag; any other `.tfvars` file requires an explicit `-var-file` flag or the `*.auto.tfvars` naming convention.
- `sensitive = true` on both variables and outputs prevents accidental exposure in terminal logs and CI build output, though it does not encrypt the value inside the state file itself.
- `TF_VAR_*` environment variables are the standard way to inject secrets in CI without ever writing them to a file on disk.
- The validation block on a variable lets you enforce constraints (valid CIDR, allowed enum values, length limits) beyond what the type system alone can express.
- Outputs form the interface between configurations when using modules (Chapter 12) — a module's outputs become the values a parent configuration references.

NOTE

Precedence order (highest to lowest) is: `-var / -var-file` command-line flags, `TF_VAR_*` environment variables, then `terraform.tfvars`, then the variable's own default.

Best Practices

- Use consistent, descriptive variable names in `snake_case` (e.g. `subnet_cidr`, not `cidr1` or `c`), and always add a description so teammates don't have to read the resource block to understand intent.
- Never commit a `terraform.tfvars` file containing real secrets to version control — add it to `.gitignore` if it holds sensitive defaults, and commit only a `terraform.tfvars.example` template instead.
- For genuine secrets (passwords, API keys, tokens), prefer Google Secret Manager over `tfvars` entirely, referencing the secret at apply time instead of storing it in any file — covered in full in Chapter 18.
- Mark any variable or output that touches credentials as `sensitive = true` as a matter of habit, even knowing it still lands in state — it is a cheap, meaningful layer of defense against accidental console exposure.

Key Takeaways

- Input variables make configurations reusable across environments; `terraform.tfvars` supplies real values without editing any `.tf` file.
- `locals` reduce repetition for values computed from other variables or resources, without being externally configurable.
- `TF_VAR_*` environment variables keep secrets out of files entirely, which is ideal for CI/CD pipelines.
- Outputs expose resource attributes after apply and chain configurations together, with `sensitive = true` hiding values from default console output.

Troubleshooting tip: if a variable seems to be silently ignored, check for a stray `-var-file` flag

in your shell history or CI script that is overriding your `terraform.tfvars` — precedence order, not a typo, is usually the culprit.

Chapter 10: Terraform Modules

As infrastructure grows, copy-pasting the same VPC, cluster, or instance configuration across environments becomes error-prone. A **module** is a reusable, self-contained package of Terraform configuration that accepts inputs and produces outputs, like a function in a programming language. Modules give you three things: DRY code (write the pattern once, call it many times), consistency (every environment — dev, staging, production — gets the exact same validated resource shape, instead of three subtly different hand-written copies that inevitably drift), and encapsulation (callers only see declared inputs and outputs, never the internal resource wiring, so the module's implementation can be refactored later without breaking anyone who calls it). The root module — the directory where you run `terraform apply` — calls child modules to compose larger systems from small, independently testable building blocks, the same way a well-factored application is built from small functions rather than one giant script.

A Reusable VPC Module

Here is a minimal but functional module that creates a custom VPC with one subnet, parameterized so it can be reused across dev, staging, and production.

```
modules/  
└─ vpc/  
    ├── main.tf  
    ├── variables.tf  
    └─ outputs.tf
```

`modules/vpc/variables.tf` declares the inputs the module accepts, including a sensible default and a validation rule so misuse fails fast at plan time rather than after resources are created:

```
variable "vpc_name" {  
  type      = string  
  description = "Name for the VPC network"  
}  
  
variable "subnet_cidr" {  
  type      = string  
  description = "CIDR range for the subnet"  
  
  validation {  
    condition     = can(cidrhost(var.subnet_cidr, 0))  
    error_message = "subnet_cidr must be a valid CIDR block."  
  }  
}  
  
variable "region" {  
  type      = string  
  description = "Region for the subnet"  
  default   = "us-central1"  
}  
  
variable "project_id" {  
  type      = string  
  description = "GCP project ID"  
}
```

`modules/vpc/main.tf` contains the actual resources:

```

resource "google_compute_network" "vpc" {
  name          = var.vpc_name
  project       = var.project_id
  auto_create_subnetworks = false
}

resource "google_compute_subnetwork" "subnet" {
  name          = "${var.vpc_name}-subnet"
  ip_cidr_range = var.subnet_cidr
  region        = var.region
  network       = google_compute_network.vpc.id
  project       = var.project_id
}

```

modules/vpc/outputs.tf exposes values the caller may need:

```

output "network_self_link" {
  value = google_compute_network.vpc.self_link
}

output "subnet_self_link" {
  value = google_compute_subnetwork.subnet.self_link
}

```

Calling the Module

From the root module, reference the module by its relative path and pass in variables:

```

module "dev_vpc" {
  source      = "../modules/vpc"
  vpc_name    = "dev-vpc"
  subnet_cidr = "10.10.0.0/24"
  region      = "us-central1"
  project_id  = "my-gcp-project-123456"
}

```

Reusing the module for another environment is now a five-line block with different values, no copy-pasted resource definitions, and no risk of the two environments drifting apart in structure. Because region has a default, a staging environment that also runs in us-central1 can omit it entirely:

```

module "staging_vpc" {
  source      = "../modules/vpc"
  vpc_name    = "staging-vpc"
  subnet_cidr = "10.20.0.0/24"
  project_id  = "my-gcp-project-123456"
}

```

Both calls instantiate the same underlying resources with different names and CIDR ranges, and both are tracked independently in state as `module.dev_vpc.*` and `module.staging_vpc.*` — a bug fix or security hardening change made once inside `modules/vpc/main.tf` automatically applies to every environment the next time `terraform apply` runs, which is the core productivity win of modularizing infrastructure.

Key points about this pattern:

- `variables.tf` is the module's public contract; only what's declared there can be set by the

caller.

- `main.tf` resource names (`google_compute_network.vpc`) are local to the module and don't collide with root-level resources of the same name.
- `outputs.tf` is the only way data flows back out; everything else stays encapsulated.
- Running `terraform init` after adding a module downloads/links it and generates a lock entry.
- Module inputs can have default values to make them optional, and validation blocks to enforce constraints (e.g., CIDR format).

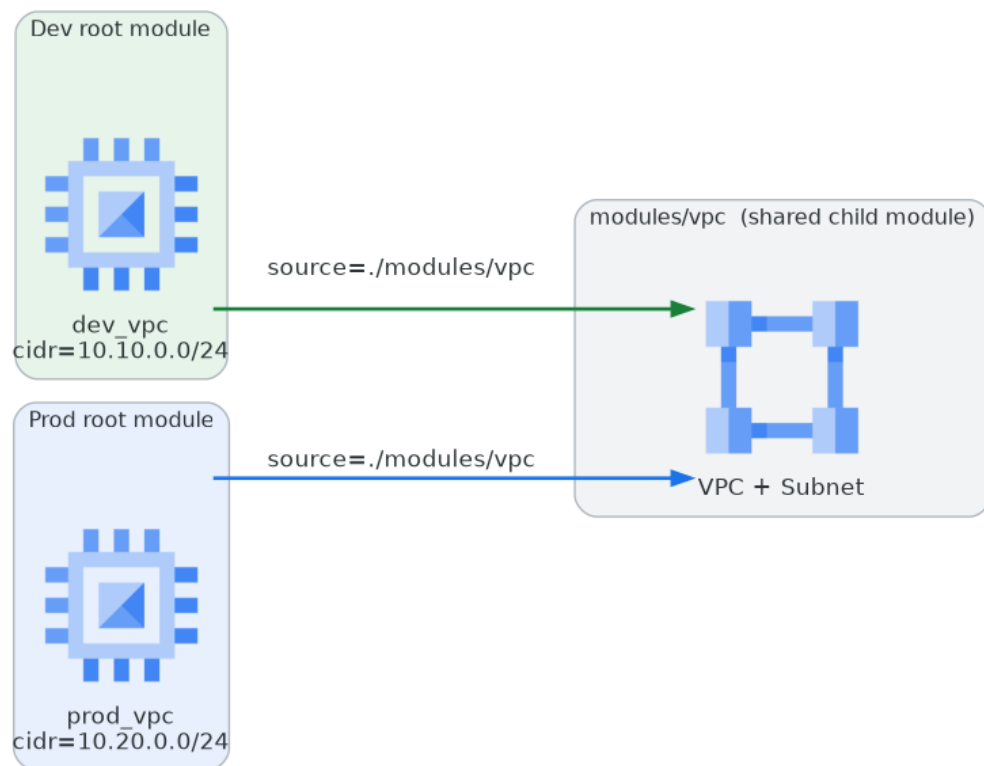


Figure 8: Two environments, one shared child module: *dev* and *prod* root modules both call `modules/vpc` with different variable values instead of duplicating resource logic.

Source Variants and Consuming Outputs

The source argument controls where Terraform fetches a module from: a local relative path (`./modules/vpc`) for in-repo modules, a Terraform Registry address (`hashicorp/network/google`) for vetted community/verified modules, or a Git URL (`git::https://github.com/org/repo.git//path?ref=v`) for private or versioned team modules. To use a module's output elsewhere in the same configuration, reference it as `module.<name>.<output_name>` — for example, `module.dev_vpc.network_self_link` can be passed directly into a `google_compute_instance` resource's `network_interface` block.

BEST PRACTICE

Pin module versions (`version = "~> 6.0"` for registry modules, or a ref tag for Git modules) so upstream changes don't silently break your environment on the next `terraform init -upgrade`.

Nesting and Organizing Modules

Modules can call other modules, so a top-level network module might internally wrap a vpc module and a firewall module, exposing only the outputs downstream consumers actually need. As a codebase grows, a common layout keeps all shared modules under a single `modules/` directory at the repository root, with per-environment root configurations (`environments/dev`, `environments/prod`) that each call the same modules with different variable values. This separates “what infrastructure looks like” (the module) from “which environment gets how much of it” (the root configuration), and it lets teams review module changes once in a pull request instead of re-reviewing near-duplicate resource blocks scattered across every environment directory.

Versioning and Testing Modules

Once a module is used by more than one root configuration, treat it like a small library: give it its own directory (or its own Git repository, for cross-team sharing), commit an example root configuration under `modules/vpc/examples/basic/` that calls it with realistic values, and run `terraform validate` and `terraform plan` against that example in CI on every change. This catches syntax errors, missing variable descriptions, and broken references before any real environment ever picks up the change. Tag releases with semantic version numbers (`v1.0.0`, `v1.1.0`) when the module lives in its own Git repository, so consumers can pin a ref and upgrade deliberately rather than being surprised by an unannounced breaking change the next time someone runs `terraform init -upgrade`. A `CHANGELOG` at the module root, even a short one, saves significant debugging time when three environments are running three different versions of the same module simultaneously.

Key Takeaways

- Modules turn repeated resource patterns into reusable, parameterized building blocks.
- A module’s `variables.tf` and `outputs.tf` form its public interface; internals stay encapsulated.
- `source` can point to a local path, the Terraform Registry, or a Git URL, each suited to different sharing needs.
- Reference module outputs with `module.<name>.<output>` to wire modules together.

Chapter 11: Managing GCP Compute Engine

Compute Engine is GCP's IaaS offering: virtual machines you fully control, billed per second, with configurable CPU, memory, disks, and networking. Unlike managed services such as Cloud Run or App Engine, Compute Engine gives you the underlying operating system, so it's the right choice whenever a workload needs custom kernel settings, licensed software with specific OS requirements, or full control over the boot process. Terraform's `google_compute_instance` resource models a VM as a single declarative block covering machine sizing, disks, networking, and boot-time configuration together, so the same configuration that provisions a dev box can provision a fleet of identical production servers just by changing a variable or wrapping the resource in `count/for_each`.

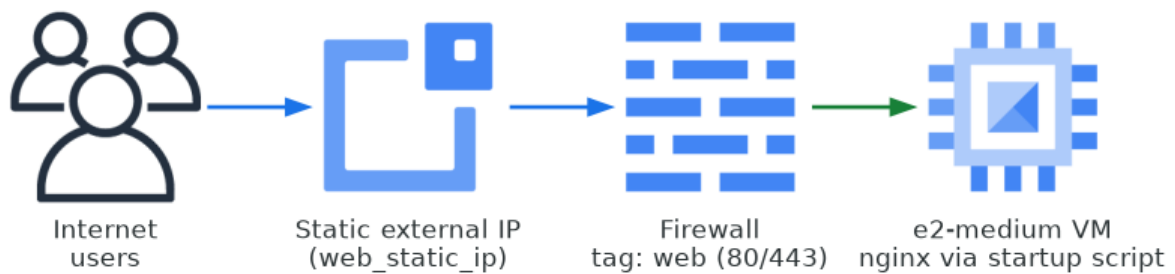


Figure 9: Traffic flow to the web server: internet users reach a static IP, pass the tag-scoped firewall rule, and land on the nginx VM.

A Complete Web Server Instance

The example below provisions one e2-medium VM running Debian 12, reserves a static external IP, opens it to HTTP/SSH traffic via network tags, installs and starts nginx on first boot, and attaches billing/organizational labels.

```
resource "google_compute_address" "web_static_ip" {
  name     = "web-static-ip"
  region   = "us-central1"
}

resource "google_compute_instance" "web_server" {
  name         = "web-server-01"
  machine_type = "e2-medium"
  zone         = "us-central1-a"

  boot_disk {
    initialize_params {
      image = "debian-cloud/debian-12"
      size  = 20
      type  = "pd-balanced"
    }
  }
}

network_interface {
```

```

    network = "default"
    access_config {
      nat_ip = google_compute_address.web_static_ip.address
    }
  }

  metadata_startup_script = <<-EOF
  #!/bin/bash
  apt-get update
  apt-get install -y nginx
  systemctl enable nginx
  systemctl start nginx
EOF

  tags = ["web", "ssh-allowed"]

  labels = {
    environment = "dev"
  }
}

output "web_server_ip" {
  value = google_compute_address.web_static_ip.address
}

```

Key Lines Explained

- `google_compute_address` is a standalone reserved static IP resource; referencing it in `access_config.nat_ip` guarantees the VM's public IP survives a rebuild (an ephemeral IP would change).
- `boot_disk.initialize_params.image` pulls the public `debian-12` image family from the `debian-cloud` project; Terraform tracks the resolved image digest to detect drift.
- `network_interface` with an `access_config` block (even empty) is what grants the VM an external IP; omitting it creates an internal-only instance.
- `metadata_startup_script` runs as root on every boot via the `google-startup-scripts` guest agent — ideal for lightweight bootstrapping like installing packages, though for anything complex a custom image or config management tool is preferable.
- `tags` are lightweight string labels used exclusively for **network targeting** — a firewall rule with `target_tags = ["web"]` applies only to instances tagged `web`.
- `labels` are key-value metadata used for **billing breakdowns, resource organization, and filtering** in Cloud Console/BigQuery billing exports; they have no effect on firewall or IAM behavior.

In short: use tags when a firewall or routing rule needs to select instances; use labels when you need to track cost or ownership. Both can be applied to the same instance and often are.

`e2-medium` (2 vCPUs, 4 GB memory) is a reasonable default for light web workloads and keeps costs predictable; heavier workloads move to `e2-standard-4` or compute-optimized `c2` machine types without changing anything else in the resource. The `pd-balanced` disk type blends SSD performance with lower cost compared to `pd-ssd`, and is the sensible default unless a workload is genuinely disk-I/O bound, in which case `pd-ssd` trades higher cost for more consistent IOPS.

Scaling to Multiple Instances

Provisioning a fleet instead of one server is a matter of adding count (or for_each for named instances) to the same resource block, rather than duplicating it:

```
resource "google_compute_instance" "web_server" {
  count      = 3
  name      = "web-server-0${count.index + 1}"
  machine_type = "e2-medium"
  zone      = "us-central1-a"
  # ...boot_disk, network_interface, metadata_startup_script, tags, labels
  ↪ unchanged
}
```

Each instance gets its own address in Terraform state (google_compute_instance.web_server[0], [1], [2]), so instances can be individually inspected, tainted, or replaced without touching the others — the same pattern used later for managed instance groups when workloads need to autoscale based on load rather than a fixed count.

Firewall Rule to Match

The web tag is only useful once a firewall rule targets it:

```
resource "google_compute_firewall" "allow_web" {
  name      = "allow-web-ssh"
  network   = "default"
  target_tags = ["web", "ssh-allowed"]
  source_ranges = ["0.0.0.0/0"]

  allow {
    protocol = "tcp"
    ports    = ["22", "80", "443"]
  }
}
```

Apply with the usual workflow:

```
terraform init
terraform plan -out=tfplan
terraform apply tfplan
```

```
google_compute_address.web_static_ip: Creating...
google_compute_instance.web_server: Creating...
google_compute_instance.web_server: Creation complete after 32s
Apply complete! Resources: 3 added, 0 changed, 0 destroyed.
```

TIP

If nginx isn't reachable after apply, SSH in and check `sudo journalctl -u google-startup-scripts.service` — it shows the exact output and exit status of your startup script, which is the fastest way to debug a failed bootstrap.

Service Accounts and Instance Identity

Beyond tags and labels, an instance's service_account block controls what GCP APIs the VM itself can call without embedding any credentials — the recommended pattern for anything the startup script or an application on the VM needs to talk to, such as reading a config file from Cloud

Storage or writing logs:

```
resource "google_compute_instance" "web_server" {
  # ...other arguments unchanged

  service_account {
    email  = google_service_account.web_sa.email
    scopes = ["cloud-platform"]
  }
}
```

Pairing a purpose-built `google_service_account` with IAM role bindings scoped to only what the instance needs (e.g., `roles/storage.objectViewer` on one bucket) is far safer than relying on the broad default Compute Engine service account, which historically carries project-editor-level access unless explicitly restricted.

Key Takeaways

- A single `google_compute_instance` resource, combined with a static `google_compute_address`, fully defines a reachable, self-configuring VM.
- `metadata_startup_script` is the simplest way to bootstrap software at boot; check its logs via `journalctl -u google-startup-scripts.service`.
- Use tags for firewall targeting and labels for cost/organizational metadata — they serve different purposes and are not interchangeable.
- Firewall rules and instances are decoupled resources linked only by matching tag strings, so plan tag names deliberately.

Chapter 12: Managing GKE with Terraform

Google Kubernetes Engine separates two concerns: the **control plane** (API server, scheduler, etcd), which Google manages, patches, and scales for you at no infrastructure burden, and **node pools**, groups of Compute Engine VMs that run your workloads and that you configure via Terraform. GKE offers two modes: **Standard**, where you manage node pool sizing, machine types, and upgrades yourself for maximum control, and **Autopilot**, where Google also manages nodes and you pay per pod resource request rather than per VM. Standard trades some operational simplicity for fine-grained control over cost and machine shape; Autopilot trades that control for near-zero node management. This chapter focuses on Standard mode since it maps most directly onto explicit Terraform resources, though the same `google_container_cluster` resource can create an Autopilot cluster simply by setting `enable_autopilot = true` and omitting node pool resources entirely.

Cluster and Node Pool

Terraform models a GKE deployment as two resources: `google_container_cluster` for the control plane and networking shell, and one or more `google_container_node_pool` resources for the actual compute.

```
resource "google_container_cluster" "primary" {
  name     = "dev-gke-cluster"
  location = "us-central1-a"
  project  = "my-gcp-project-123456"

  network     = "default"
  subnetwork  = "default"

  remove_default_node_pool = true
  initial_node_count       = 1

  ip_allocation_policy {
    cluster_secondary_range_name = "pods-range"
    services_secondary_range_name = "services-range"
  }
}

resource "google_container_node_pool" "primary_nodes" {
  name       = "primary-node-pool"
  cluster    = google_container_cluster.primary.name
  location   = "us-central1-a"
  project    = "my-gcp-project-123456"

  node_count = 1

  node_config {
    machine_type = "e2-medium"
    disk_size_gb = 50
    disk_type    = "pd-balanced"
    oauth_scopes = ["https://www.googleapis.com/auth/cloud-platform"]
  }

  autoscaling {
    min_node_count = 1
    max_node_count = 5
  }
}
```

```
}  
}
```

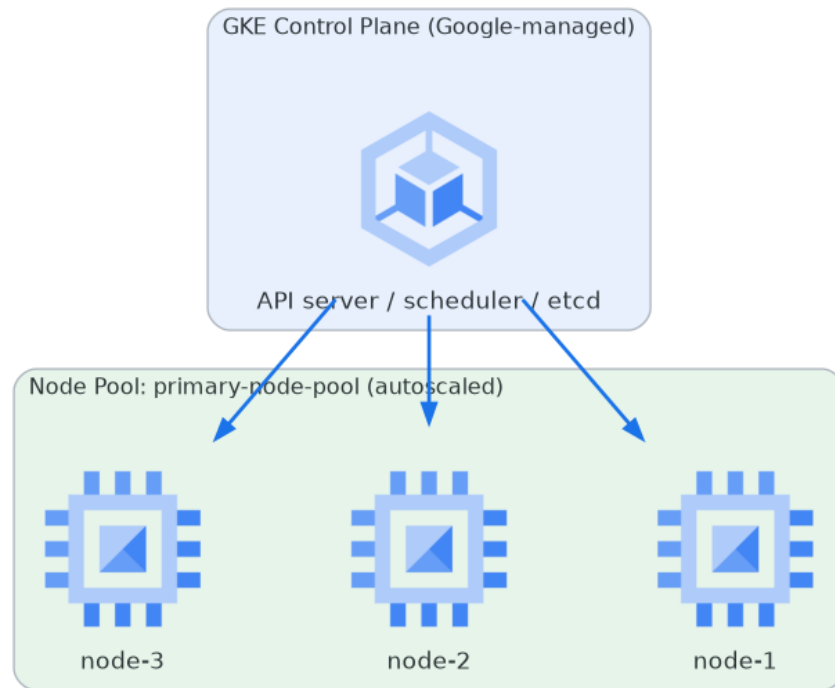


Figure 10: Google manages the GKE control plane; Terraform manages the autoscaled node pool beneath it.

Why Remove the Default Node Pool

Creating a `google_container_cluster` always provisions a default node pool internally, but that pool cannot be customized through Terraform's node pool arguments (machine type, autoscaling, labels) after the fact — it's tied to cluster creation. Setting `remove_default_node_pool = true` with `initial_node_count = 1` lets Terraform spin up the cluster with a throwaway single-node pool, then immediately delete it, leaving you to define real node pools explicitly and fully manage them (size, machine type, autoscaling, upgrades) as separate, independently-scalable resources.

VPC-Native Clusters and Secondary Ranges

The `ip_allocation_policy` block makes the cluster **VPC-native** rather than routes-based: pods and services get IP addresses from dedicated secondary ranges (`pods-range`, `services-range`) carved out of the VPC subnet ahead of time, rather than relying on custom routes. This gives pods routable, firewall-addressable IPs directly within the VPC, which is required for many GCP integrations (Cloud Armor, VPC Service Controls, Private Service Connect) and is the default/recommended mode for all new clusters.

Apply and connect with `kubectl`:

```
terraform apply -auto-approve  
gcloud container clusters get-credentials dev-gke-cluster \  
  --zone us-central1-a --project my-gcp-project-123456
```

```
kubectl get nodes
```

Release Channels and Versioning

GKE clusters subscribe to a **release channel** (RAPID, REGULAR, or STABLE) rather than pinning an exact Kubernetes version by default, which lets Google handle patch and minor version upgrades automatically on a cadence matching your risk tolerance:

```
resource "google_container_cluster" "primary" {
  # ...other arguments unchanged
  release_channel {
    channel = "REGULAR"
  }
}
```

STABLE receives the most heavily tested versions latest, RAPID gets new Kubernetes features soonest, and REGULAR is the balanced default for most production workloads. Pinning an exact `min_master_version` instead of using a channel is possible but shifts the burden of tracking Kubernetes end-of-life dates and manually triggering upgrades entirely onto your team.

Autoscaling and Node Pool Notes

- `node_count` on the pool sets a fixed baseline; the autoscaling block lets the Cluster Autoscaler grow/shrink between `min_node_count` and `max_node_count` based on pending pod demand.
- `oauth_scopes` controls what GCP APIs nodes can call by default; scope this down in production and prefer Workload Identity for pod-level permissions instead of broad node scopes.
- Node pools can be destroyed and recreated independently of the cluster, enabling rolling machine-type or disk changes with zero control-plane disruption.
- Multiple node pools (e.g., a spot-pool with preemptible VMs alongside `primary-node-pool`) are common for mixing cost-optimized and stable capacity.
- `management` and `upgrade_settings` blocks on the node pool control whether GKE auto-repairs unhealthy nodes and how surge upgrades roll out new node versions with minimal workload disruption.

Applying Workloads

Once `kubectl` is pointed at the cluster, standard Kubernetes manifests work unmodified — Terraform manages the cluster and node infrastructure, while application deployment typically stays in `kubectl apply` or a GitOps tool, though the `kubernetes` and `helm` Terraform providers can also manage in-cluster resources declaratively if a single tool is preferred end-to-end:

```
kubectl create deployment nginx --image=nginx:latest
kubectl expose deployment nginx --port=80 --type=LoadBalancer
kubectl get service nginx
```

The `LoadBalancer` service type automatically provisions a GCP external Network Load Balancer, demonstrating how GKE resources transparently create underlying GCP infrastructure outside of Terraform's direct view — a good reason to keep cluster/node-pool infrastructure in Terraform while treating in-cluster load balancer services as day-to-day operational objects.

BEST PRACTICE

For production, enable a private cluster (`private_cluster_config { enable_private_nodes = true }`) so nodes have no public IPs, and enable Workload Identity

```
(workload_identity_config { workload_pool = "my-gcp-project-123456.svc.id.goog"
}) so pods authenticate to GCP APIs via scoped Kubernetes service account bindings instead
of shared node-level credentials.
```

Key Takeaways

- Google manages the GKE control plane; Terraform manages node pools, which are the actual billable, configurable compute.
- Always pair `remove_default_node_pool = true` with an explicit `google_container_node_pool` to get full control over machine type, autoscaling, and lifecycle.
- `ip_allocation_policy` with secondary ranges makes a cluster VPC-native, giving pods and services routable GCP-integrated IPs.
- Production clusters should combine private nodes with Workload Identity to minimize exposed attack surface and avoid broad node-level credentials.

Chapter 13: Managing Cloud Storage

Cloud Storage buckets are one of the most frequently Terraform-managed resources in a GCP estate: application logs, Terraform state itself, data-pipeline staging areas, and static website assets all typically land in a bucket. Provisioning them through code means every bucket gets the same naming convention, the same retention policy, and the same IAM review, instead of drifting configuration created ad hoc through the console. This chapter walks through a single consolidated example that covers the settings you will reach for on nearly every production bucket: a globally unique name, a location and storage class, object versioning, cost-driven lifecycle rules, and a scoped IAM grant.

A Complete Bucket Example

```
resource "google_storage_bucket" "app_data" {
  name           = "my-gcp-project-123456-app-data"
  project        = "my-gcp-project-123456"
  location       = "US"
  storage_class  = "STANDARD"

  uniform_bucket_level_access = true

  versioning {
    enabled = true
  }

  lifecycle_rule {
    condition {
      age = 30
    }
    action {
      type          = "SetStorageClass"
      storage_class = "NEARLINE"
    }
  }

  lifecycle_rule {
    condition {
      age = 365
    }
    action {
      type = "Delete"
    }
  }

  labels = {
    environment = "production"
    team        = "platform"
  }
}

resource "google_storage_bucket_iam_member" "viewer" {
  bucket = google_storage_bucket.app_data.name
  role   = "roles/storage.objectViewer"
  member = "group:data-analysts@example.com"
}
```

```
}
```

How It Works

- name must be globally unique across every GCP project on the planet, so production configurations typically embed the project ID as a prefix to avoid collisions with other customers' buckets.
- location accepts a specific region (us-central1), a dual-region, or a multi-region value like US; multi-region buckets cost slightly more per GB but replicate data across multiple regions automatically, which matters for disaster-recovery-sensitive workloads.
- storage_class sets the default class new objects land in; STANDARD is appropriate for actively read/written data, while NEARLINE, COLDLINE, and ARCHIVE progressively trade retrieval latency and per-operation cost for lower storage cost.
- versioning { enabled = true } keeps prior versions of an object when it is overwritten or deleted, which is cheap insurance against an application bug or an operator mistake that clobbers important files.
- The first lifecycle_rule block matches any object older than 30 days and transitions it to Nearline storage, automatically shrinking storage cost for data that is rarely re-read after its first month. The second block matches objects older than 365 days and deletes them outright, enforcing a one-year retention ceiling without any manual cleanup job.
- google_storage_bucket_iam_member grants exactly one role to exactly one principal, additively, leaving any other bindings already present on the bucket untouched. Prefer it over google_storage_bucket_iam_binding, which instead replaces the entire membership list for a role and can silently revoke access granted by another team's Terraform run or a manual console change.

BEST PRACTICE

Always set `uniform_bucket_level_access = true` on new buckets. It disables legacy per-object ACLs in favor of IAM-only permission checks, which is far simpler to audit, easier to reason about during a security review, and is Google's recommended default going forward.

Lifecycle rules compose freely: GCP evaluates each `lifecycle_rule` block independently against every object in the bucket, so a single bucket can simultaneously downgrade storage class at 30 days and enforce hard deletion at 365 days, exactly as shown above. For buckets holding regulated or compliance-sensitive data, consider layering in a `retention_policy` block to prevent deletion before a fixed minimum duration, and pair versioning with a lifecycle rule keyed on `condition.num_newer_versions` to cap how many noncurrent versions accumulate, which otherwise grows storage cost unbounded over time.

Common Pitfalls

A few mistakes show up repeatedly in real Cloud Storage configurations, and each is easy to avoid once you know to look for it:

- **Forgetting force_destroy.** By default, Terraform refuses to delete a bucket that still contains objects. Set `force_destroy = true` only on genuinely disposable buckets (scratch, CI artifacts); leave it unset on anything holding production data so a `destroy` fails loudly instead of quietly deleting real content.
- **Choosing the wrong location for cost.** A multi-region bucket that only ever serves traffic from us-central1 pays for cross-region replication it never uses; pin `location` to the single region your workload actually runs in unless you have a genuine multi-region access pattern.
- **Skipping lifecycle rules entirely.** Buckets without any lifecycle policy grow forever at STANDARD pricing. Even a simple age-based transition to Nearline or Coldline, as shown above, often cuts storage cost substantially for logs and backups that are rarely re-read

after the first few weeks.

- **Granting IAM roles at the project level instead of the bucket level.** `roles/storage.objectViewer` granted on the project gives read access to every bucket in it. Scoping the same role to a single bucket, as the `google_storage_bucket_iam_member` example does, keeps the blast radius of a compromised credential much smaller.

Terraform's `google_storage_bucket` resource also supports `cors`, `website`, and `encryption` blocks for static hosting and customer-managed encryption keys, but the versioning, lifecycle, and IAM combination above covers the overwhelming majority of day-to-day bucket configurations you will write.

Key Takeaways

- One `google_storage_bucket` resource can combine versioning with multiple `lifecycle_rule` blocks to balance durability against storage cost automatically.
- `google_storage_bucket_iam_member` is the safe, additive way to grant per-bucket roles such as `roles/storage.objectViewer` without disturbing other bindings.
- Prefer uniform bucket-level access over legacy per-object ACLs for simpler, IAM-centralized permission management across every bucket in the project.

Chapter 14: Managing Cloud SQL

Cloud SQL provisioning through Terraform ties together a database instance, a database-level user, and project IAM so that application workloads can connect securely without embedding static credentials anywhere in source control. Getting the instance settings right up front — availability type, backup strategy, and network exposure — matters more here than for most GCP resources, because some of them (like the database engine version) cannot be changed later without recreating the instance. This chapter walks through one consolidated, production-ready example and the reasoning behind each setting.

Architecture: App to Cloud SQL

Applications should never connect to Cloud SQL directly over its public IP address. Instead, the Cloud SQL Auth Proxy runs alongside the application (as a sidecar container or a local process) and handles encrypted, IAM-authenticated connections to a regional instance that maintains an automatic standby replica for failover:

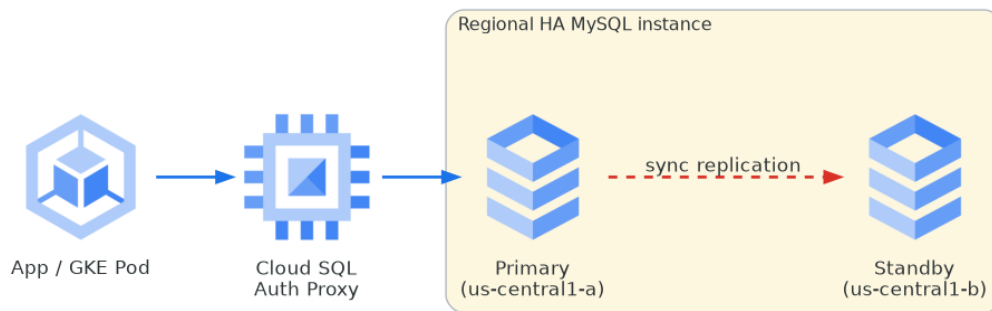


Figure 11: Applications never talk to Cloud SQL directly — the Auth Proxy handles encrypted, IAM-authenticated connections to a regional HA instance.

A Complete Cloud SQL Example

```
resource "google_sql_database_instance" "main" {
  name            = "app-db-primary"
  project         = "my-gcp-project-123456"
  region         = "us-central1"
  database_version = "MYSQL_8_0"

  deletion_protection = true

  settings {
    tier                = "db-custom-2-8192"
    availability_type   = "REGIONAL"

    backup_configuration {
      enabled                = true
      point_in_time_recovery_enabled = true
      start_time             = "02:00"
      transaction_log_retention_days = 7
    }
  }
}
```

```

    ip_configuration {
      ipv4_enabled = false
      private_network = "projects/my-gcp-project-123456/global/networks/default"
    }
  }
}

resource "google_sql_user" "app_user" {
  name      = "app_service"
  instance = google_sql_database_instance.main.name
  password = var.db_password
}

resource "google_project_iam_member" "sql_client" {
  project = "my-gcp-project-123456"
  role    = "roles/cloudsql.client"
  member  =
↪ "serviceAccount:app-runtime@my-gcp-project-123456.iam.gserviceaccount.com"
}

```

How It Works

- `database_version` pins both the engine and its major version; changing this field later generally requires standing up a new instance and migrating data, rather than an in-place upgrade, so pick it deliberately.
- `tier` selects the machine shape backing the instance; custom tiers like `db-custom-2-8192` (2 vCPUs, 8 GB RAM) let you size precisely instead of choosing from a fixed shared-core or standard tier list.
- `availability_type = "REGIONAL"` provisions a synchronous standby replica in a second zone within the region, giving automatic failover during a zone outage with no data loss for committed transactions. Omit it, or set it to `ZONAL`, only for genuinely non-critical or development workloads where downtime is acceptable.
- `backup_configuration` with `point_in_time_recovery_enabled = true` turns on binary logging (MySQL) or WAL archiving (PostgreSQL), which lets you restore to any specific second within the retention window rather than being limited to the timestamp of the last nightly snapshot.
- `ip_configuration.ipv4_enabled = false` disables the instance's public IP entirely; paired with `private_network`, all traffic stays inside the VPC, and the Auth Proxy (or Private Service Connect) is the only supported path in.
- `google_sql_user.password` should be sourced from a Terraform variable backed by Secret Manager or a CI/CD secret store — never hardcoded as a literal string in version-controlled configuration.
- The `roles/cloudsql.client` binding is what allows the Auth Proxy sidecar to authenticate as the given service account and open a connection; it grants no SQL-level privileges by itself, since those are managed separately inside the database engine with standard `GRANT` statements.

WARNING

Always set `deletion_protection = true` on production Cloud SQL instances. Without it, a stray terraform `destroy`, an accidental removal of the resource block, or a careless `-target` apply will delete the instance — and every byte of its data — with no recovery path beyond whatever backups you had configured.

Common Pitfalls

Several mistakes recur often enough in production Cloud SQL configurations to call out explicitly:

- **Choosing ZONAL availability for a customer-facing database.** A single-zone instance has no automatic failover, so a zone outage means downtime until Google restores the zone or an operator manually promotes a replica. REGIONAL availability costs roughly double the compute, but for anything customers depend on, that is usually a reasonable trade.
- **Leaving the public IP enabled by default.** Terraform's provider defaults `ipv4_enabled` to true. Explicitly setting it to false, as shown above, forces all traffic through the private network path and removes an entire class of exposure to internet-based attacks.
- **Sizing backups by snapshot count instead of retention days.** `transaction_log_retention_days` controls how far back point-in-time recovery can reach; a value of 1 or 2 days is common in default provider examples but is often too short for real incident response, which can take days to notice and diagnose.
- **Managing database-level grants outside Terraform without documenting them.** `google_sql_user` creates the login, but privileges on individual schemas or tables are typically applied with SQL GRANT statements. Track those in a versioned SQL migration file alongside the Terraform configuration so the two never drift out of sync.

Cloud SQL also supports read replicas via a second `google_sql_database_instance` resource with `master_instance_name` set, which is worth reaching for once a single primary's read load becomes a bottleneck, rather than scaling the primary's tier indefinitely.

Key Takeaways

- A single `google_sql_database_instance` resource can combine regional high availability with point-in-time recovery backups in one block.
- The Cloud SQL Auth Proxy plus a `roles/cloudsql.client` IAM binding is the standard, credential-free way for applications to reach a private-IP instance.
- Set `deletion_protection = true` in every production configuration to guard against accidental, irreversible data loss.

Chapter 15: IAM with Terraform

GCP's IAM model rests on three interlocking ideas. A **role** is a named bundle of permissions, ranging from broad built-in roles like `roles/editor` down to a handful of specific actions. A **member** is whoever the role is granted to — a human user, a Google group, or a service account — and is what actually shows up in a policy binding. A **service account** is a non-human identity used by applications, CI pipelines, and Terraform itself to authenticate against GCP APIs without a human's credentials. When the built-in roles are too broad for a given job, a **custom role** lets you assemble the exact set of permissions a workload actually needs, nothing more. Terraform manages all four of these as ordinary resources, but the resource type you choose for granting a role matters just as much as which role you pick.

Granting a Role Additively

```
resource "google_project_iam_member" "logging_writer" {
  project = "my-gcp-project-123456"
  role    = "roles/logging.logWriter"
  member  =
    ↪ "serviceAccount:app-runtime@my-gcp-project-123456.iam.gserviceaccount.com"
}
```

`google_project_iam_member` adds exactly one member to exactly one role, without touching any other bindings already present for that same role. That makes it safe to run alongside other teams' Terraform configurations, alongside manually granted console changes, and alongside other modules in the same project that also happen to reference `roles/logging.logWriter`.

WARNING

`google_project_iam_binding` is authoritative, not additive — applying it replaces the *entire* membership list for a role with only the members listed in that one resource block. If another team, a different Terraform config, or a console click had previously granted that same role to someone else, an authoritative binding will silently remove them on the next apply. Reserve `google_project_iam_binding` for roles you are certain your configuration fully owns, and default to `google_project_iam_member` everywhere else.

Defining a Custom Role

```
resource "google_project_iam_custom_role" "storage_lister" {
  role_id    = "storageLister"
  project    = "my-gcp-project-123456"
  title      = "Storage Bucket Lister"
  description = "Read-only access to list buckets and objects"

  permissions = [
    "storage.buckets.list",
    "storage.buckets.get",
    "storage.objects.list",
  ]
}

resource "google_project_iam_member" "grant_custom_role" {
  project = "my-gcp-project-123456"
  role    = google_project_iam_custom_role.storage_lister.id
  member  = "group:audit-team@example.com"
}
```

```
}
```

- `role_id` becomes part of the role's resource name and must be unique within the project; it cannot be reused even after the role is deleted, so pick something descriptive from the start.
- `permissions` is an explicit allow-list, not an inherited set — start as narrow as the workload will tolerate, and add individual permissions only as real “permission denied” errors surface in practice, rather than copying the permission list from the nearest built-in role.
- `title` and `description` are purely cosmetic but show up in the IAM console, so a clear description saves the next engineer from having to reverse-engineer intent from a permissions list.
- Custom roles are referenced by other IAM resources through their `.id` output, exactly the same way a built-in role string like `"roles/viewer"` would be used, which keeps the granting resource identical regardless of role source.

Service accounts (`google_service_account`) are typically created once, independently, and then referenced by `member = "serviceAccount:${google_service_account.x.email}"` inside whichever IAM binding resources need to grant it access. Keeping identity creation and permission granting as two separate resource types — rather than one combined block — makes each easier to review, and lets the same service account be granted different roles across several unrelated Terraform configurations without duplication.

A Third Option: Conditional and Audit-Friendly Grants

Beyond `_member` and `_binding`, most IAM resources also accept an optional `condition` block, which restricts when a grant is active — for example, limiting a role to expire on a fixed date, or scoping it to resources matching a specific expression. This is particularly useful for temporary elevated access, such as granting `roles/owner` to an on-call engineer for a 4-hour incident window instead of leaving a standing broad grant in place indefinitely. Combined with Cloud Audit Logs, which record every IAM policy change and every API call made under a given identity, conditional grants let you keep permissions narrow by default and widen them only for a bounded, auditable window.

A related but distinct question is where in the resource hierarchy a grant should live. Roles applied at the organization or folder level cascade down to every project underneath, which is efficient for genuinely org-wide roles (like a security team's read access to all logs) but dangerous for anything narrower, since a mistake at that level affects every project at once. As a rule of thumb, grant roles at the lowest level in the hierarchy — project, then resource — that satisfies the actual requirement, and reserve organization- and folder-level bindings for roles that are genuinely meant to apply everywhere.

Key Takeaways

- Use `google_project_iam_member` for additive, safe-to-coexist role grants; reserve `google_project_iam_binding` only for roles your configuration fully and exclusively owns.
- `google_project_iam_custom_role` lets you define a minimal, explicit permission set rather than relying on the broader reach of built-in roles.
- Treat service account creation and IAM role granting as separate resources so identities and their permissions can evolve and be reviewed independently.

Chapter 16: Terraform Workspaces

Terraform workspaces let a single configuration directory manage multiple isolated state files — commonly one per environment such as `dev`, `qa`, and `prod` — by switching a name that other expressions can read at runtime through `terraform.workspace`. The appeal is obvious: no duplicated `.tf` files across environments, and one `terraform plan` command works everywhere. The trade-off is just as real, though — every workspace shares the exact same variable definitions, provider version, and resource logic, so a bug in the configuration hits every environment identically, and a mistaken `terraform destroy` executed in the wrong workspace still reaches real, live infrastructure through the same code path with no structural barrier to stop it. For that reason, most teams managing production infrastructure prefer a directory-per-environment layout instead — separate `dev/`, `qa/`, and `prod/` directories, each with its own backend configuration, variable values, and often distinct provider credentials — because it enforces genuinely stronger separation of state and blast radius than a shared workspace ever can. Workspaces remain a good fit for short-lived, low-risk environments: feature-branch preview stacks, ephemeral integration-test environments, or personal sandboxes that get created and destroyed frequently.

Working with Workspaces

```
terraform workspace list
terraform workspace new dev
terraform workspace select dev
terraform workspace show
```

```
default
* dev
```

Created and switched to workspace "dev"!

```
dev
```

`terraform workspace list` shows every workspace that currently exists for the configuration, with an asterisk marking the active one. `new` creates a workspace and switches into it immediately, while `select` switches between workspaces that already exist without creating anything new. `show` simply prints the name of whichever workspace is currently active, which is worth running before any `apply` or `destroy` as a sanity check. Each workspace's state is stored separately within the same backend — for the `gcs` backend, as a distinct object under a workspace-prefixed path in the same bucket — so switching workspaces never risks reading or writing another environment's state file by accident, even though the configuration files themselves are shared.

Using `terraform.workspace` in Configuration

```
locals {
  instance_size = {
    default = "e2-micro"
    dev     = "e2-micro"
    qa      = "e2-medium"
    prod    = "e2-standard-4"
  }
}

resource "google_compute_instance" "app" {
  name      = "app-${terraform.workspace}"
  project   = "my-gcp-project-123456"
```

```

zone          = "us-central1-a"
machine_type  = local.instance_size[terraform.workspace]

boot_disk {
  initialize_params {
    image = "debian-cloud/debian-12"
  }
}

network_interface {
  network = "default"
}
}

```

The `locals.instance_size` map keys directly on `terraform.workspace`, so simply switching workspaces automatically resizes the instance without touching any resource block: `dev` provisions a minimal `e2-micro`, `qa` steps up to `e2-medium`, and `prod` gets a larger `e2-standard-4`. The same lookup pattern extends naturally to instance counts, backup retention windows, or any other environment-scoped value, keeping a single configuration DRY across environments that otherwise share an identical resource shape. Note that `terraform.workspace` also flows nicely into resource naming, as shown in the `name` attribute above, which keeps environment-specific resources easy to distinguish in the console.

TIP

Always confirm the active workspace with `terraform workspace show` immediately before running `apply` or `destroy` — it is surprisingly easy to assume you are still in `dev` when a previous terminal session left the workspace switched to `prod`.

When Workspaces Still Make Sense

None of this means workspaces are never the right tool. They shine when every environment truly is a disposable, short-lived copy of the same stack — a preview environment spun up per pull request and torn down on merge, a load-testing sandbox created for an afternoon, or a personal development copy an individual engineer creates and destroys at will. In those cases the shared-configuration trade-off that makes workspaces risky for production is exactly what makes them convenient elsewhere: one `terraform apply` after switching workspaces is faster than maintaining a full parallel directory, and there is no meaningful “wrong environment” risk when every environment is equally disposable.

A hybrid approach is also common in practice: use directory-per-environment for the handful of environments that matter (staging and production), while using workspaces underneath a single `ci/` or `preview/` directory to multiply out as many ephemeral, matching-shaped environments as pull requests demand. This keeps the blast radius of production isolated structurally, while still getting the DRY benefit of workspaces where the risk of a mistake is low. Whichever approach you choose, keep the decision consistent across a given repository — mixing workspace-based and directory-based patterns for the same environment tier tends to confuse whoever runs `terraform apply` next, since the mental model for “which environment am I about to affect” no longer has one single answer.

Key Takeaways

- Workspaces isolate state per environment within one configuration but share code, variables, and provider setup — a real risk once production is involved.
- Directory-per-environment is the safer default for production isolation; reserve workspaces for low-risk, ephemeral, or feature-branch environments instead.

- `terraform.workspace` can drive a `locals` lookup map to vary machine sizing, counts, or naming per environment without duplicating resource blocks.

Chapter 17: CI/CD Integration

Running Terraform from a laptop does not scale for teams. Local applies leave no audit trail, depend on whichever provider versions happen to be cached on that machine, and give no one else visibility into what changed and why. CI/CD pipelines solve all three problems: they execute Terraform consistently from a clean environment, they record every plan and apply as a build log tied to a commit, and they force a review step before infrastructure changes reach GCP. This chapter walks through the same core pattern implemented in four common platforms — GitHub Actions, GitLab CI, Azure DevOps, and Jenkins — so you can recognize the pattern regardless of which tool your organization already uses.

The Core Pattern

Every platform in this chapter implements the same lifecycle. `terraform init` always runs first against a shared remote backend so state is never local to the runner. `terraform plan` runs on a pull or merge request so reviewers can see the exact diff before anything is applied — this is the single most valuable habit in this chapter, since it turns infrastructure changes into a reviewable artifact just like application code. `terraform apply` runs only after the request is merged, and ideally only after a human explicitly approves that specific apply, not just the code review that preceded it.

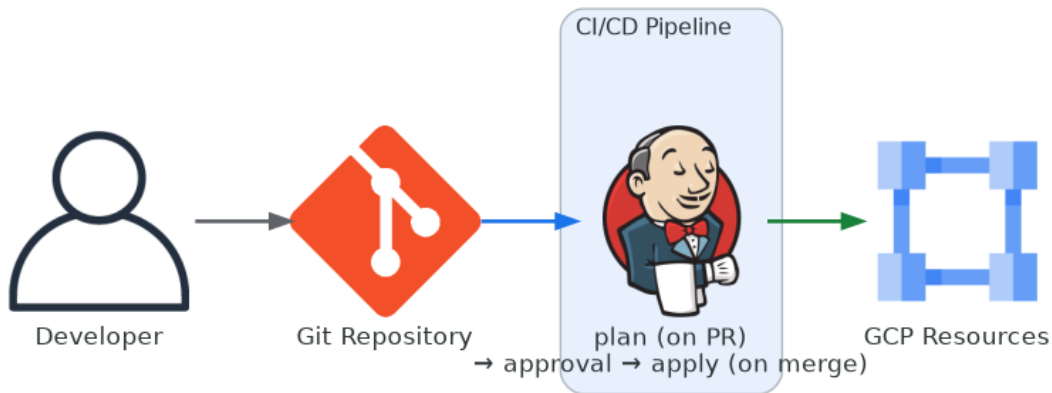


Figure 12: The shared CI/CD pattern: every platform in this chapter plans on the PR, waits for approval, and only applies after merge.

Keeping plan and apply as separate, clearly labeled pipeline stages — rather than one combined step — is what makes this pattern auditable: anyone can look at the pipeline history and see exactly which plan output a given apply corresponds to.

GitHub Actions

GitHub Actions uses the community hashicorp/setup-terraform action to install the CLI, and workflow triggers (`pull_request` versus `push` to main) to separate the plan stage from the apply stage. The plan job's output can also be posted back to the PR as a comment using the action's `terraform_wrapper` output, giving reviewers the diff without leaving GitHub.

```
name: terraform
on:
  pull_request:
```

```

push:
  branches: [main]
jobs:
  terraform:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: hashicorp/setup-terraform@v3
      - run: terraform init
      - run: terraform plan -out=tfplan
      - if: github.ref == 'refs/heads/main'
        run: terraform apply -auto-approve tfplan

```

GitLab CI

GitLab CI expresses the same idea with named stages and the built-in `merge_request_event` rule, which restricts the plan job to running only when a merge request is open. GitLab also understands Terraform's JSON plan format natively and renders it as a readable diff directly inside the merge request UI when the plan step emits a `.tfplan.json` report artifact.

```

stages: [validate, plan, apply]
plan:
  stage: plan
  script:
    - terraform init
    - terraform plan -out=tfplan
  rules:
    - if: $CI_PIPELINE_SOURCE == "merge_request_event"
apply:
  stage: apply
  script: terraform apply -auto-approve tfplan
  rules:
    - if: $CI_COMMIT_BRANCH == "main"
  when: manual

```

The `when: manual` directive on the apply job is what forces a human to click a button in the GitLab UI rather than letting a merge to main trigger an apply automatically.

Azure DevOps

Azure Pipelines uses the community `TerraformTaskV4` task and multi-stage YAML. Its distinguishing feature is the environment resource, which lets you attach approval checks (specific reviewers, business-hours restrictions, and so on) directly to a stage in the Azure DevOps portal rather than encoding the approval logic in YAML itself.

```

stages:
  - stage: Plan
    jobs:
      - job: Plan
        steps:
          - task: TerraformTaskV4@4
            inputs: {command: 'init'}
          - task: TerraformTaskV4@4
            inputs: {command: 'plan'}
  - stage: Apply

```

```

jobs:
- deployment: Apply
  environment: 'production'
  strategy:
    runOnce:
      deploy:
        steps:
          - task: TerraformTaskV4@4
            inputs: {command: 'apply'}

```

Jenkins

A declarative Jenkinsfile achieves the same separation with an input step, which pauses the pipeline and waits for a human to click “proceed” in the Jenkins UI before the apply stage runs — Jenkins’ equivalent of a required reviewer gate, without relying on any platform-native approval feature.

```

pipeline {
  agent any
  stages {
    stage('Plan') {
      steps { sh 'terraform init && terraform plan -out=tfplan' }
    }
    stage('Approval') {
      steps { input message: 'Apply to GCP?' }
    }
    stage('Apply') {
      steps { sh 'terraform apply -auto-approve tfplan' }
    }
  }
}

```

Authenticating CI to GCP

All four platforms should authenticate as a dedicated service account, and that authentication should use Workload Identity Federation rather than a downloaded JSON key whenever the platform supports it (GitHub Actions, GitLab, and Azure DevOps all do). Workload Identity Federation exchanges a short-lived, platform-issued token for a GCP access token at runtime, so no long-lived credential is ever stored as a pipeline secret that could leak in a log or a misconfigured job. Grant that service account only the roles needed for the resources the pipeline manages — see Chapter 18 for detailed least-privilege guidance — rather than reusing a broadly scoped account across every pipeline in the organization.

BEST PRACTICE

Never let a pipeline auto-apply to production without a manual approval step, and always use a remote backend with state locking (a GCS backend with native locking, or a Terraform Cloud workspace) so two concurrent pipeline runs can never corrupt the same state file.

Multi-Environment Pipelines and Drift Detection

Most organizations run the same pipeline against multiple environments — dev, staging, and prod — rather than one pipeline per environment. The cleanest way to do this is a separate state file (and often a separate backend prefix or workspace) per environment, with the environment name passed into the pipeline as a variable that selects the right `.tfvars` file. This keeps a bug

in a dev-only variable from ever touching prod state, and it lets you promote a known-good plan from staging to prod without re-running `terraform plan` against different inputs than what was actually reviewed.

It is also worth scheduling a periodic `terraform plan` run — nightly or weekly, outside of any code change — purely to detect drift: manual console changes, resources modified outside Terraform, or expired credentials that would otherwise only surface the next time someone tries to make a real change. A non-empty plan from this scheduled run should alert the team, since it means the deployed environment no longer matches the configuration in source control, even though no one intentionally changed anything.

Key Takeaways

- All CI/CD platforms follow the same plan-on-PR, apply-on-merge pattern regardless of syntax differences between them.
- GitHub Actions and GitLab CI use trigger conditions (`pull_request` vs `push`) to separate plan and apply into distinct jobs.
- Azure DevOps environments and Jenkins input steps provide built-in manual approval gates before a production apply runs.
- Authenticate CI runners with Workload Identity Federation and least-privilege service accounts, and always use a locked remote state backend.

Chapter 18: Security Best Practices

Terraform manages the infrastructure that everything else runs on, so a mistake here — a leaked credential, an over-permissioned service account, an unencrypted state file sitting in a public bucket — can compromise an entire environment in one step, not just one resource. Because Terraform has broad create/update/delete authority by design, the security posture of your Terraform setup deserves at least as much scrutiny as the applications it deploys. This chapter covers the four control areas every GCP Terraform project should have in place before it reaches production: secrets management, state file security, key rotation, and least-privilege service accounts.

Secrets Management

Never hardcode secrets — database passwords, API keys, TLS private keys — in `.tf` files or as variable defaults, and never commit a `.tfvars` file that contains them. Store secrets in Secret Manager instead, and reference them from a data source at apply time, so the secret's value never appears in your Terraform source code and is only pulled into memory when a resource actually needs it.

```
resource "google_secret_manager_secret" "db_password" {
  secret_id = "db-password"
  replication { auto {} }
}

resource "google_secret_manager_secret_version" "db_password_v1" {
  secret      = google_secret_manager_secret.db_password.id
  secret_data = var.db_password
}

data "google_secret_manager_secret_version" "db_password" {
  secret = google_secret_manager_secret.db_password.id
}
```

Downstream resources reference `data.google_secret_manager_secret_version.db_password.secret_data` so the plaintext value is fetched fresh from Secret Manager rather than being duplicated across multiple resource blocks. Note that the initial value passed into `secret_data` still flows through Terraform state during the apply that creates the version, which is exactly why state itself must be protected as described next — Secret Manager reduces where a secret lives in your source code, but it does not remove state from the threat model.

State File Security

Terraform state files routinely contain plaintext secrets, connection strings, and complete metadata for every managed resource, which makes state one of the most sensitive artifacts in your entire infrastructure — arguably more sensitive than the resources it describes, since a single state file can expose secrets for dozens of resources at once. Three controls matter most in practice. First, store state in a GCS bucket encrypted with a Customer-Managed Encryption Key (CMEK) rather than relying on Google-managed encryption alone, so your organization controls the key and can revoke it independently of Google. Second, enable object versioning on that bucket, so a corrupted, maliciously overwritten, or accidentally deleted state file can be rolled back to a known-good version rather than being lost outright. Third, restrict IAM on the state bucket as tightly as possible — grant `roles/storage.objectAdmin` scoped to that single bucket to only the specific service accounts and engineers who genuinely need to run Terraform against it, never a project-wide storage role. Enable object-level audit logging on the bucket as well, so every state read and write is traceable to a specific identity and timestamp, which matters when investigating an unexpected change.

Key Rotation

Long-lived service account JSON keys are the single most common source of GCP credential leaks in practice, because they are static, easy to accidentally commit to a repository, and typically never expire on their own unless someone remembers to rotate them. The strongest fix is to avoid them entirely: prefer Workload Identity Federation for both CI/CD pipelines and human access, so short-lived tokens are issued on demand for each session instead of a static key being distributed to a laptop or build server where it can be copied or leaked. Where a static secret genuinely cannot be avoided — a third-party API key with no federation support, for example — store it in Secret Manager and rotate it by adding a new secret version on a defined schedule, updating the consuming Terraform data source reference to pull the latest version, and disabling (not immediately deleting) the prior version. Disabling first gives any in-flight process using the old value a short grace period before it stops working entirely, which avoids an unplanned outage during rotation.

Least Privilege and Service Account Security

Every service account that Terraform creates or authenticates as should be scoped to the minimum set of roles its workload actually requires, using predefined roles like `roles/cloud-sql.client` or a custom role rather than broad ones like `roles/editor` or `roles/owner`, which grant far more access than almost any automation genuinely needs. Avoid attaching keys to the default compute service account and disable it where it is not in active use, since it is a common target precisely because it exists by default in every project. Prefer resource-scoped bindings such as `google_service_account_iam_member` over project-level `google_project_iam_member` bindings wherever the provider supports it, since a resource-scoped binding limits the blast radius if that identity is ever compromised. Finally, run IAM Recommender or Policy Analyzer checks periodically, since permission scope tends to expand over time as teams copy existing bindings for convenience rather than granting exactly what a new workload needs.

WARNING

A single project-level `roles/owner` binding on a CI service account defeats every other control in this chapter — audit IAM bindings on your Terraform service accounts as carefully as you audit the resources they manage.

Provider-Level Safeguards

A few provider-level settings reinforce everything above without adding real complexity. Pin the google provider to a specific version range in your `required_providers` block so an unexpected provider upgrade cannot silently change default behavior for a security-sensitive resource. Set `impersonate_service_account` on the provider block when a human needs to run Terraform locally against a project, rather than distributing that service account's key file directly to their workstation — the human authenticates with their own identity and is granted short-lived impersonation rights instead. Finally, treat `terraform plan` output itself as sensitive in CI logs: mark any pipeline variables containing secret values so the platform automatically redacts them, since a plan that references a secret's value can otherwise leak it into a build log that far more people can read than the underlying Secret Manager entry.

Enterprise Recommendations

- Store all Terraform state in CMEK-encrypted, versioned GCS buckets with narrowly scoped, bucket-level IAM.
- Use Workload Identity Federation instead of downloaded service account keys wherever the CI/CD platform and use case support it.
- Keep all secrets in Secret Manager, referenced via data sources, never as plain variables or in `.tfvars` committed to git.

- Grant Terraform service accounts least-privilege, resource-scoped roles, and review those bindings on a recurring quarterly cadence.
- Enable audit logging on both the state bucket and Secret Manager so every access is traceable to a specific identity.

Key Takeaways

- Reference secrets from Secret Manager via a data source rather than hardcoding them anywhere in Terraform configuration.
- Encrypt state with CMEK, enable bucket versioning, and restrict IAM to the minimum set of principals who need it.
- Prefer Workload Identity Federation over long-lived service account keys; rotate unavoidable secrets via Secret Manager versions.
- Apply least privilege to every Terraform-managed service account and audit its bindings on a regular schedule.

Chapter 19: Advanced Terraform Concepts

Beyond basic resource blocks, Terraform provides several constructs that make configurations more flexible, more reusable, and safer to change over time. This chapter walks through six of the most useful for GCP work, each with a short working example: `dynamic` blocks, `for_each`, `count`, `depends_on`, `lifecycle`, and conditional expressions. None of these are exotic — you will use most of them in nearly every real module you write — but they are easy to misuse if you have not seen the common pitfalls.

Dynamic Blocks

A dynamic block generates repeated nested blocks — like the `allow` rules inside a `firewall` resource — from a list or map, instead of writing each one out by hand as a separate static block.

```
resource "google_compute_firewall" "web" {
  name      = "allow-web"
  network   = google_compute_network.vpc.id

  dynamic "allow" {
    for_each = ["80", "443"]
    content {
      protocol = "tcp"
      ports    = [allow.value]
    }
  }
}
```

This is most useful when the number of nested blocks varies by environment or input — the `for_each` list can come from a variable instead of a hardcoded literal, so a dev environment with one open port and a prod environment with five do not require two different resource definitions.

for_each

`for_each` creates one resource instance per key in a map or set, keyed by that value rather than by numeric position, so adding or removing an entry only affects the specific instance tied to that key.

```
resource "google_storage_bucket" "env" {
  for_each = toset(["dev", "staging", "prod"])
  name     = "my-gcp-project-123456-${each.key}"
  location = "us-central1"
}
```

Removing `"staging"` from the set destroys only that one bucket; `dev` and `prod` are untouched because each is tracked internally by its own key (`each.key`), not by its position in a list — this is the key advantage `for_each` has over `count` for any collection that might change shape later.

count

`count` creates a fixed number of near-identical resource copies, indexed numerically through `count.index`, and is simpler than `for_each` when you genuinely just need `N` identical copies of something.

```
resource "google_compute_instance" "worker" {
  count      = 3
  name      = "worker-${count.index}"
}
```

```

machine_type = "e2-medium"
zone         = "us-central1-a"
}

```

The pitfall to remember: because each instance is tracked by its numeric index rather than a stable key, removing an early element — say `worker-0` — shifts every subsequent instance's index down by one, and Terraform will destroy and recreate resources that did not actually change just to realign the indices. Use `for_each` instead whenever the underlying collection can shrink, grow, or reorder over the resource's lifetime; reserve count for genuinely fixed-size, order-independent sets.

depends_on

Terraform infers most ordering automatically from attribute references between resources, but `depends_on` forces an explicit ordering when a dependency exists that isn't expressed through any attribute reference — most commonly, an API needing to be enabled before a resource that relies on it can be created.

```

resource "google_project_service" "sql" {
  service = "sqladmin.googleapis.com"
}

resource "google_sql_database_instance" "main" {
  name         = "primary-db"
  depends_on = [google_project_service.sql]
}

```

Use `depends_on` sparingly — reach for it only when Terraform's automatic reference-based ordering genuinely cannot see the dependency, since overusing it makes a module's real dependency graph harder to read from the code alone. A good test before adding it: if the dependency could instead be expressed by referencing an attribute of the upstream resource (even one you do not otherwise need, such as `google_project_service.sql.id`), prefer that reference over an explicit `depends_on` entry, since it keeps the ordering visible directly in the resource's arguments.

lifecycle

The `lifecycle` block changes how Terraform manages a resource's create, update, and destroy behavior. `prevent_destroy` is the most commonly used safeguard, and it is particularly valuable on stateful, hard-to-recreate resources like production databases where an accidental destroy would mean real data loss.

```

resource "google_sql_database_instance" "prod" {
  name           = "prod-db"
  database_version = "POSTGRES_15"

  lifecycle {
    prevent_destroy = true
  }
}

```

With this set, any plan that would delete `prod-db` — including an accidental terraform destroy, a resource rename that Terraform interprets as delete-and-recreate, or a change to an argument that forces replacement — fails outright instead of proceeding, forcing a deliberate, explicit override before the resource can actually be removed.

Conditional Expressions

Terraform's ternary syntax, `condition ? true_val : false_val`, lets a single expression pick between two values based on a variable, which is commonly used to size or configure resources differently per environment without duplicating entire resource blocks.

```
resource "google_compute_instance" "app" {
  machine_type = var.environment == "prod" ? "n2-standard-4" : "e2-small"
  name         = "app-server"
  zone         = "us-central1-a"
}
```

TIP

Combine conditionals with `for_each` and `lookup()` against an environment-keyed map of settings rather than chaining several nested ternaries together, which quickly become difficult to read and debug as more environments are added.

Combining These Techniques

These six constructs are rarely used in isolation in a real module. A typical GCP module might use `for_each` to create one instance per environment key, a dynamic block inside each instance to attach a variable number of network tags, a ternary to pick machine type per environment, and a lifecycle block to protect the production instance specifically — all inside the same resource. Reading Terraform code fluently means recognizing each of these patterns individually so that a module combining several of them at once is still easy to reason about rather than looking like unrelated special cases stacked on top of each other.

When reviewing someone else's module, it is worth checking specifically for two anti-patterns this chapter's examples are meant to prevent: `count` used on a collection that is expected to shrink or reorder over its lifetime (a `for_each` should have been used instead), and a missing `prevent_destroy` on any resource holding data that cannot be trivially regenerated, such as a production database or a bucket containing customer uploads. Both are quick to spot once you know to look for them, and both have caused real production incidents when missed.

Key Takeaways

- Use dynamic blocks to generate repeated nested blocks from a variable-driven list instead of duplicating them by hand for each environment.
- Prefer `for_each` over `count` whenever a collection can shrink, grow, or reorder, since index-based count resources shift and get recreated unnecessarily.
- Use `depends_on` only for dependencies Terraform cannot infer, and `lifecycle { prevent_destroy = true }` to guard critical stateful resources like production databases.
- Ternary conditional expressions provide clean, readable per-environment logic without duplicating entire resource blocks.

Chapter 20: Terraform Enterprise Architecture

Running Terraform at enterprise scale is less about clever HCL and more about process: how changes are proposed, reviewed, and promoted safely across many teams, dozens of projects, and multiple environments. A single engineer running `terraform apply` from a laptop does not scale past a handful of people, and it removes every audit trail an enterprise needs. This chapter covers the workflow, resource hierarchy, and drift-detection patterns that keep large GCP estates consistent as the number of contributors and environments grows.

The GitOps Pipeline

Enterprise Terraform adopts a GitOps model: the Git repository is the single source of truth, and no one applies changes directly. A change enters through a feature branch, is validated by CI, reviewed by a human, and only then promoted to production by an automated pipeline—never by a person typing `apply`.

Every box in that pipeline exists to answer a specific audit question later: who proposed this change, who approved it, what exactly did it do, and can we prove it matched policy before it touched production.

Branching and Review Culture

Infrastructure changes deserve the same rigor as application code: small, focused pull requests, a `terraform plan` posted automatically as a PR comment so reviewers see the actual diff before approving, and mandatory review from at least one other engineer—often the platform team specifically for cross-cutting modules like VPC, IAM, or org policy. CI authenticates to GCP through Workload Identity Federation rather than long-lived service account keys, which eliminates one of the most common credential-leak vectors in enterprise pipelines. Policy-as-code tools—Sentinel in HCP Terraform, or OPA/Conftest wired into a custom pipeline—can automatically block a PR that violates a guardrail, for example disallowing publicly-accessible Cloud SQL instances or requiring cost-center labels on every resource before merge is even possible.

Ownership matters as much as process. Large repositories typically use a CODEOWNERS file so that a change touching the shared VPC module automatically requests review from the network team, while a change scoped to one application’s own environment folder only needs sign-off from that team. This keeps review load proportional to blast radius: nobody wants every application team waiting on the platform team to approve a routine variable bump, but nobody wants an application team silently reshaping shared networking either.

Drift Detection

Live infrastructure inevitably drifts from state: someone edits a firewall rule in the console during an incident, or a GCP-managed field changes on its own. A scheduled job running `terraform plan -detailed-exitcode` on a cron—hourly or nightly, depending on how sensitive the environment is—catches this before it causes a production surprise. Exit code 2 means changes are pending against the current configuration; wiring that signal into Slack or PagerDuty turns silent drift into a same-day ticket instead of a mystery someone discovers weeks later during an unrelated `apply`.

```
terraform plan -detailed-exitcode -out=drift.tfplan
# 0 = no changes, 1 = error, 2 = drift detected
```

The alert should include enough context to triage quickly—which environment, which module, and a link to the plan output—so an on-call engineer can tell within seconds whether the drift is benign (a GCP-managed label update) or the early sign of an out-of-band change that needs to be reverted or reconciled into the configuration.

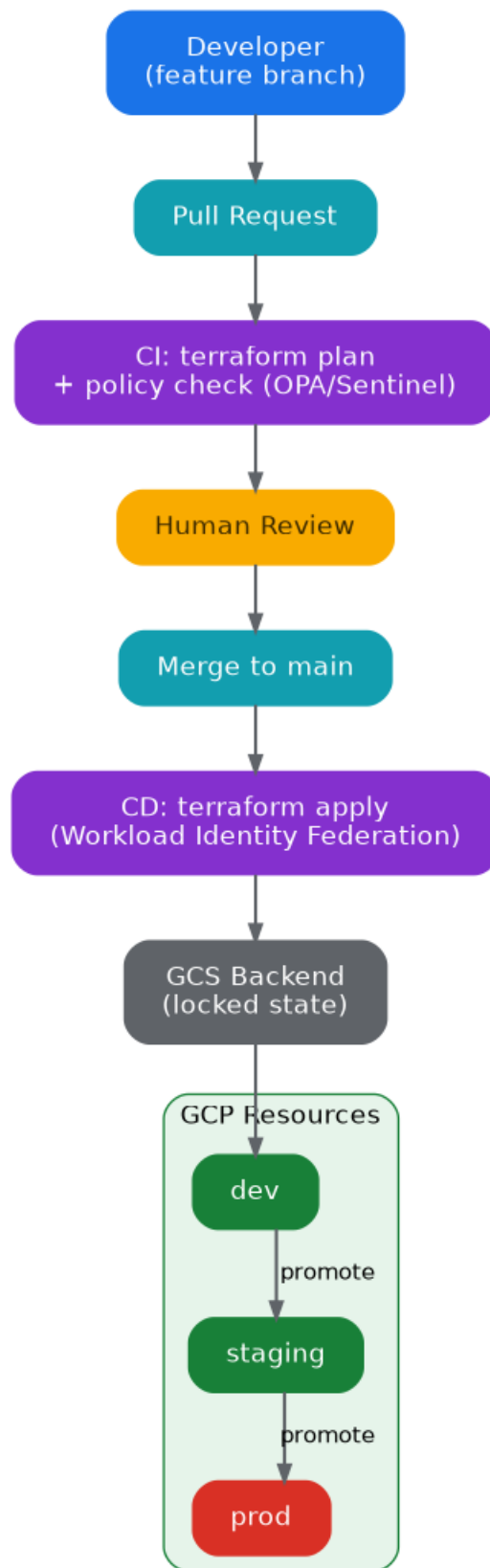


Figure 13: The enterprise GitOps pipeline: every change flows through CI plan, policy check, human review, and a keyless CD apply before reaching dev, staging and prod in turn.

Promoting Modules Across Environments

Environments should consume the *same* module source pinned to deliberately different, advancing versions—never copy-pasted HCL duplicated per environment.

```
module "vpc" {  
  source = "git::https://source.example.com/modules/vpc.git?ref=v2.3.0"  
  project = var.project_id  
  region = "us-central1"  
}
```

dev tracks the newest tag first; once it has been validated there, staging and then prod bump their ref to that same version in turn. This guarantees the exact code that was tested in staging is byte-for-byte what ships to production, and a rollback is simply reverting one ref string rather than reconstructing what changed.

This module-version-bump pattern also gives platform teams a clean deprecation path: when a module needs a breaking change, they cut a new major version, let dev and staging adopt it on their own schedule, and only remove the old version once every environment has moved off it. Nobody is forced onto a new module version by someone else's unrelated change.

GCP Resource Hierarchy at Scale

Large organizations mirror their org chart directly in GCP's resource hierarchy: **Organization -> Folders -> Projects -> Resources**. Folders typically map to business units or environment tiers (Folder: Production, Folder: Non-Production, Folder: Shared-Services); IAM bindings and organization policy constraints set at a folder are inherited by every project beneath it, so a single policy change protects dozens of teams at once. Each application or team usually gets its own project, both for blast-radius isolation—one team's mistake can't touch another's resources—and for clean billing attribution. Terraform manages this hierarchy itself using `google_folder` and `google_project` resources, typically from a dedicated "bootstrap" configuration whose access is even more tightly restricted than application-level Terraform.

Treat the bootstrap/org-level Terraform configuration as more sensitive than any application workload—a mistake there can create or destroy entire projects, not just individual resources.

Org policy constraints layered on top of this hierarchy—restricting public IP creation, requiring OS Login, or pinning allowed regions—are set once at a folder and inherited everywhere beneath it, which is what lets a security team enforce a rule across five hundred projects with a single Terraform resource instead of five hundred manual checks.

HCP Terraform as an Alternative

Rather than hand-rolling all of the CI/CD wiring above—runners, state locking, policy checks, approval gates—many enterprises instead adopt HCP Terraform (formerly Terraform Cloud) or self-hosted Terraform Enterprise, both of which provide this run pipeline, remote execution, and governance as a managed product out of the box, trading some infrastructure-team build effort for a subscription cost and a more opinionated workflow.

Key Takeaways

- Treat infrastructure changes like application code: feature branches, CI-generated plans surfaced in the PR, and mandatory peer review before merge.
- Authenticate CI/CD with Workload Identity Federation, never static service account keys, and enforce policy-as-code checks before a merge is even allowed.
- Detect drift with scheduled `terraform plan -detailed-exitcode` runs, and promote one

- pinned module version through dev, staging, and prod rather than duplicating configuration.
- Mirror your org chart with Org -> Folders -> Projects for inherited policy and blast-radius isolation, and consider HCP Terraform/Terraform Enterprise if you don't want to build this pipeline yourself.

Chapter 21: Troubleshooting Guide

Even well-written Terraform configurations hit friction against real GCP accounts—expired credentials, unenabled APIs, permission gaps, and stuck state locks. None of these are Terraform bugs; they are the normal seams between a declarative tool and a live cloud account that changes underneath it, and they show up regardless of how carefully the HCL itself was written. This chapter is a quick-reference for the five errors you will see most often in day-to-day GCP work, with the root cause and a concrete fix for each, so you can spend less time guessing and more time shipping. Keep it near your terminal; in practice these five categories account for the overwhelming majority of support requests on any Terraform-on-GCP team.

Authentication Failures

```
Error: could not find default credentials. See
https://cloud.google.com/docs/authentication/external/set-up-adc for more
↪ information
```

Root cause: the Google provider authenticates using Application Default Credentials (ADC), and this shell, container, or CI runner has never had ADC configured—or a previous token expired and was never refreshed. This is the single most common first-run error for anyone new to a machine or a fresh CI environment. **Fix:** on a workstation, generate user ADC interactively:

```
gcloud auth application-default login
```

For a service account instead of a personal login, point `GOOGLE_APPLICATION_CREDENTIALS` at a key file, though in CI/CD you should prefer Workload Identity Federation over either approach so no long-lived credential ever touches disk. Verify the active identity at any time with `gcloud auth list` and `gcloud config list`, and double-check that the project argument in your provider block actually matches the project the credentials were issued against—a mismatch here produces a confusingly similar error even when authentication itself succeeded.

API Not Enabled

```
Error: Error creating instance: googleapi: Error 403: Compute Engine API
has not been used in project my-gcp-project-123456 before or it is disabled
```

Root cause: GCP APIs are opt-in per project; a brand-new project or one scoped down for cost control simply hasn't had the required API turned on, so Terraform's create call is rejected before it ever reaches the resource logic. **Fix:** enable the specific API the error names:

```
gcloud services enable compute.googleapis.com --project=my-gcp-project-123456
```

If you hit this repeatedly across a stack (Compute, SQL Admin, Container, Storage), it is a sign the project's API surface was never declared in Terraform. Manage it with `google_project_service` resources instead, so `terraform apply` enables everything the configuration needs up front rather than failing resource-by-resource. Note that enabling an API can itself take a minute or two to fully propagate, so an immediate retry right after `gcloud services enable` can still fail once before it succeeds—rerunning `apply` a second time is normal, not a sign of a deeper problem.

IAM Permission Errors

```
Error: Error creating instance: googleapi: Error 403: Required
↪ 'compute.instances.create'
permission for 'projects/my-gcp-project-123456'
```

Root cause: the identity running Terraform—your user account or a service account—lacks an IAM role that grants the named permission, or the correct binding was just applied and has not finished propagating (IAM changes are eventually consistent, typically settling within about 60 seconds, occasionally longer). **Fix:** confirm the binding actually exists for the right member:

```
gcloud projects get-iam-policy my-gcp-project-123456 \
  --flatten="bindings[].members" \
  --filter="bindings.members:serviceAccount:tf-runner@my-gcp-project-
  ↪ 123456.iam.gserviceaccount.com"
```

If the binding is missing, add the minimal role that covers the permission (e.g., `roles/compute.instanceAdmin.v1`) rather than reaching for `roles/owner`. If the binding is present and correct, simply wait a minute and retry before assuming something else is wrong. It's also worth checking whether the permission is being denied by an organization policy constraint rather than a missing IAM role—those produce a similarly worded 403 but require changing the policy, not the binding, so read the full error body before you start editing IAM.

State Lock Conflicts

Error: Error acquiring the state lock

Lock Info:

ID: 1a2b3c4d-5e6f-7890-abcd-ef1234567890
Operation: OpApply
Who: ci-runner@build-42

Root cause: the GCS backend uses a lock object to serialize writes to state, and either another plan/apply is genuinely running right now, or a previous run crashed, was killed, or lost network connectivity before it could release the lock cleanly. **Fix:** first confirm no other process is actually mid-run (check your CI dashboard), then release the stale lock using the ID from the error:

```
terraform force-unlock 1a2b3c4d-5e6f-7890-abcd-ef1234567890
```

Only force-unlock after you have confirmed no other apply is genuinely in progress—breaking the lock on a truly active run can let two writers corrupt the same state file simultaneously.

In CI specifically, this most often happens when a build is manually cancelled mid-apply: the job dies but never gets the chance to run its cleanup step. Adding a timeout and a guaranteed unlock-on-cancel step to the pipeline prevents most of these from ever reaching a human.

Resource Dependency Issues

Error: Cycle: google_compute_instance.app, google_compute_firewall.allow_app, google_compute_instance.app

Root cause: Terraform builds a directed graph of resources from the references in your configuration, and somewhere two or more resources reference each other's attributes, so the graph cannot be ordered—Terraform genuinely does not know which one to create first. **Fix:** break the cycle by removing the implicit two-way reference. Either add an explicit `depends_on` to force one direction, or refactor so the firewall rule keys off a static value (like a tag or CIDR) instead of an attribute the instance only produces after it exists. This is the more common fix in practice: firewall rules that target instances by network tag rather than by instance self-link avoid the cycle entirely, since tags are known before either resource is created. When the cycle isn't obvious from the error alone, render the full dependency graph to see exactly which edge loops back:

```
terraform graph | dot -Tpng > graph.png
```

Going Deeper

When these five patterns don't explain what you're seeing, turn on verbose provider logging. `TF_LOG=DEBUG terraform apply` prints every request Terraform sends to the Google APIs and every response it gets back, which is almost always enough to pinpoint the exact call GCP is rejecting and why. Redirect it to a file with `TF_LOG_PATH=debug.log` so the output survives past your scrollback, and remember to strip or redact it before sharing—debug logs frequently contain project IDs, resource names, and sometimes tokens.

Key Takeaways

- Most real-world Terraform-on-GCP errors fall into five buckets: authentication, disabled APIs, IAM propagation, state locks, and dependency cycles.
- Fix the root cause, not just the symptom—declare enabled APIs and IAM bindings in Terraform itself so they stop recurring.
- Treat `force-unlock` as a last resort, and always confirm no other run is genuinely active before using it.
- `TF_LOG=DEBUG` is your escalation path once the plain error message stops being enough to diagnose the problem.

Chapter 22: Real-World Production Project

This capstone ties together every pattern from the book into one production system: a containerized web application on GKE, backed by Cloud SQL, fronted by a global load balancer, with storage, monitoring, and a promotion pipeline across dev, staging, and prod. It is deliberately the same shape as the smaller examples earlier in the book—environments calling shared modules—scaled up to something you could actually run a business on. Nothing here is exotic; the value is in how the familiar pieces are composed, versioned, and promoted together as one coherent system rather than as a pile of independently managed resources.

Architecture Overview

A single VPC hosts three purpose-scoped subnets: one for GKE nodes, one for the database using private services access, and one reserved for management and bastion access. External traffic enters through a global HTTP(S) Load Balancer, which routes to a GKE Ingress, which in turn fans out to application Pods running across multiple zones. Pods read and write to a private Cloud SQL instance and to a Cloud Storage bucket for user uploads, while Cloud Monitoring watches request latency and error rates across the whole stack and fires alerts when either budget is breached.

Every component in this diagram is provisioned by a Terraform module, and every environment—dev, staging, prod—instantiates the same set of modules with different sizing and redundancy settings. Nothing in dev is architecturally different from prod; the only differences are node counts, machine types, and whether high-availability options are switched on, which keeps the staging environment a genuinely trustworthy rehearsal of production rather than a smaller, different system. The management subnet exists specifically so operators and CI runners can reach the cluster’s private control plane and the database without ever exposing either to the public internet, which keeps the attack surface limited to the load balancer’s public frontend alone.

Recommended Folder Structure

Environments are thin wrappers that call shared, versioned modules; no environment ever redefines resource logic on its own.

```
repo/  
  environments/  
    dev/      (backend.tf, main.tf, terraform.tfvars)  
    staging/   (backend.tf, main.tf, terraform.tfvars)  
    prod/     (backend.tf, main.tf, terraform.tfvars)  
  modules/  
    vpc/ gke/ cloudsql/ lb/ storage/ monitoring/
```

Each `environments/<env>/main.tf` sets `project`, `region`, instance sizing, and the module version pins for that environment; all real logic lives exactly once, inside `modules/`. A change to how the VPC is built—adding a new subnet, tightening a firewall rule—is written once in `modules/vpc` and picked up by every environment as it bumps its pinned version, exactly as described in the promotion workflow from the previous chapter.

Each environment also carries its own `backend.tf` pointing at a dedicated GCS bucket and state prefix, so a mistake in dev’s Terraform run can never lock or corrupt staging or prod’s state—the blast radius of any single `apply` is bounded to the one environment it targets. Keeping environments in separate GCP projects reinforces the same boundary at the IAM and billing level, not just at the state level.

Terraform Code

The three blocks below are the representative core of the stack; the load balancer, storage, and monitoring modules follow the same style and are summarized in prose afterward rather than

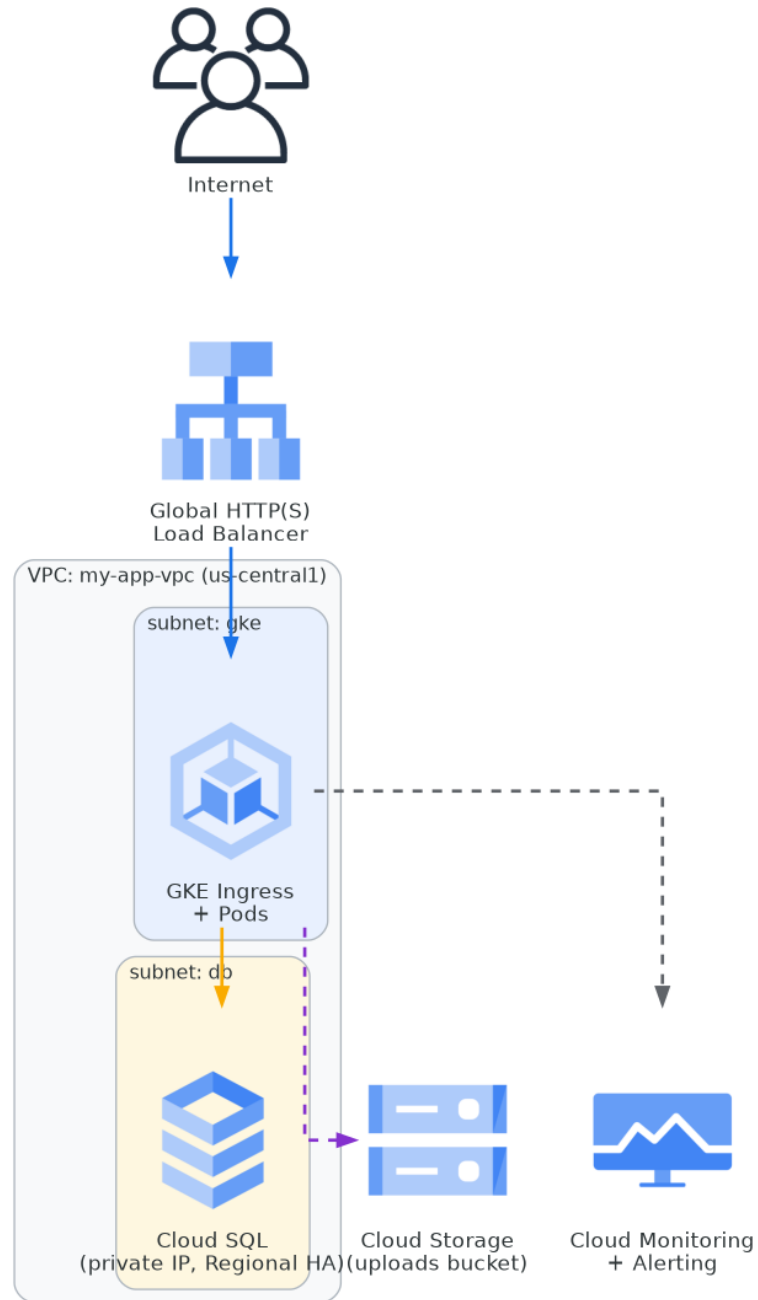


Figure 14: The capstone architecture: Internet traffic reaches GKE Ingress through a global load balancer, with Cloud SQL, Cloud Storage, and Cloud Monitoring completing the stack inside the VPC.

listed in full.

VPC and subnets. One network with subnets scoped to a single purpose each, plus secondary IP ranges GKE needs for its own Pod and Service networking:

```
resource "google_compute_network" "main" {
  name          = "my-app-vpc"
  auto_create_subnetworks = false
}

resource "google_compute_subnetwork" "gke" {
  name          = "gke-subnet"
  network       = google_compute_network.main.id
  ip_cidr_range = "10.10.0.0/20"
  region        = "us-central1"

  secondary_ip_range {
    range_name = "pods"
    ip_cidr_range = "10.20.0.0/16"
  }
  secondary_ip_range {
    range_name = "services"
    ip_cidr_range = "10.30.0.0/20"
  }
}
```

GKE cluster and node pool. A regional cluster with the default node pool removed immediately, replaced by an explicitly managed pool that autoscales with demand:

```
resource "google_container_cluster" "primary" {
  name          = "app-cluster"
  location      = "us-central1"
  network       = google_compute_network.main.id
  subnetwork    = google_compute_subnetwork.gke.id
  remove_default_node_pool = true
  initial_node_count = 1
}

resource "google_container_node_pool" "default" {
  name          = "default-pool"
  cluster       = google_container_cluster.primary.name
  location      = "us-central1"

  autoscaling {
    min_node_count = 2
    max_node_count = 6
  }
  node_config {
    machine_type = "e2-standard-4"
  }
}
```

Cloud SQL. A private Postgres instance whose tier and availability are driven entirely by an environment variable, so dev stays cheap while prod gets real redundancy:

```
resource "google_sql_database_instance" "main" {
  name = "app-db-${var.environment}"
```

```

database_version = "POSTGRES_15"
region           = "us-central1"

settings {
  tier                = var.db_tier
  availability_type   = var.environment == "prod" ? "REGIONAL" : "ZONAL"
  backup_configuration {
    enabled                = true
    point_in_time_recovery_enabled = true
  }
  ip_configuration {
    ipv4_enabled = false
    private_network = google_compute_network.main.id
  }
}
deletion_protection = var.environment == "prod"
}

```

Firewall rules, service accounts, and the management-subnet bastion follow this same compact style and are omitted here for brevity—each is a handful of resources reusing patterns already shown elsewhere in this book, not a new concept. The rest of the stack follows the same pattern without needing a full listing here. The **load balancer** module wires a `google_compute_global_forwarding_rule` to the GKE Ingress-managed backend service (`modules/lb/main.tf`), fronted by a managed SSL certificate so HTTPS termination is handled entirely by Google rather than in-cluster. The **uploads bucket** is a single `google_storage_bucket` with uniform bucket-level access and object versioning enabled (`modules/storage/main.tf`), with a lifecycle rule that moves objects older than 90 days to Nearline storage to keep costs down. And a baseline **alert policy** watches the load balancer's 5xx rate via `google_monitoring_alert_policy`, paging on-call when it crosses threshold (`modules/monitoring/main.tf`); additional policies typically cover Cloud SQL CPU utilization and GKE node pool autoscaling saturation.

Deployment Steps

The steps below are intentionally sequential and gated—each one only proceeds once the previous environment has proven itself, which is what actually prevents an untested change from reaching customers.

1. Bootstrap the GCS state bucket for each environment and enable object versioning—one-time, done before any `init`.
2. `terraform init` inside `environments/dev` to configure the remote backend.
3. `terraform plan` for dev, review the diff carefully, then `terraform apply`.
4. Run smoke tests against the dev Ingress IP: a health-check hit and one representative API call.
5. Repeat `plan/apply` for staging, pinned to the exact same module versions validated in dev.
6. Run the fuller regression and load-test suite against staging before touching prod.
7. Promote to prod: bump module refs to the validated version, get sign-off on the plan, and apply during a low-traffic window.

Each promotion step should be traceable back to a single pull request and its CI-generated plan, exactly as described for the enterprise pipeline earlier in this book—there is no separate “production process” bolted on at the end, only the same reviewed, automated pipeline pointed at a more sensitive environment.

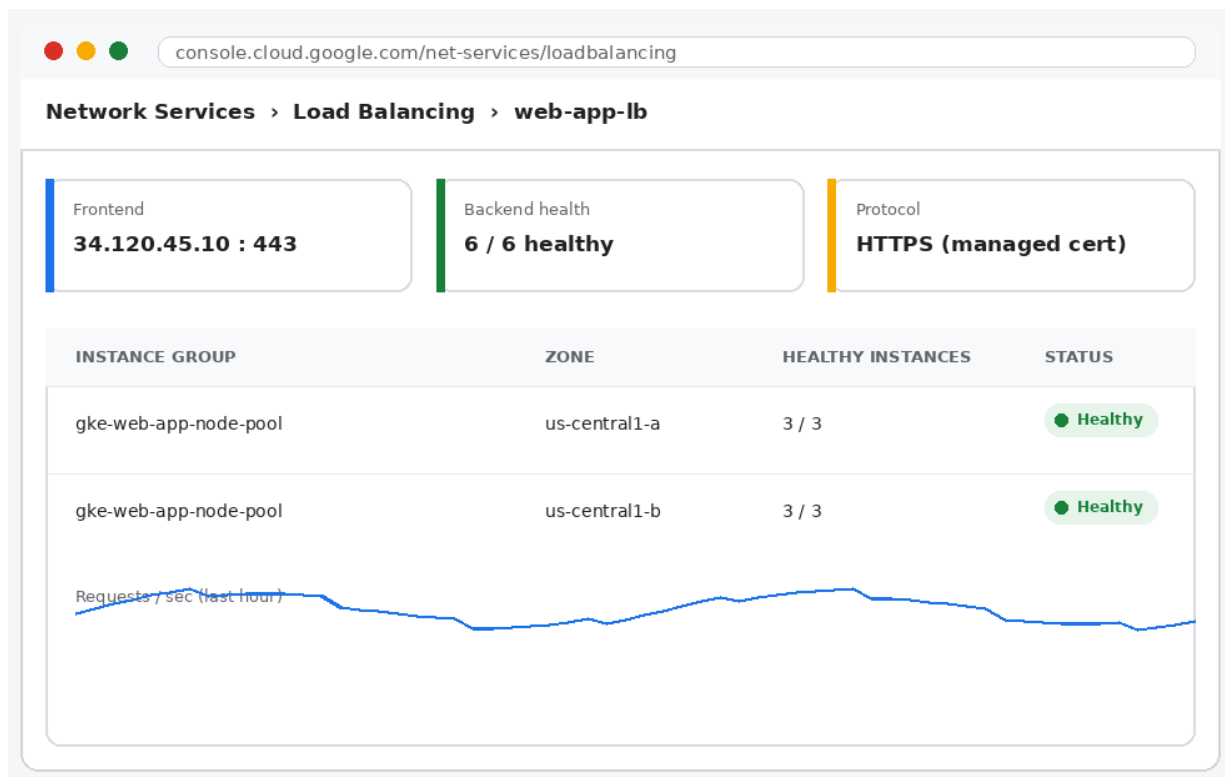


Figure 15: Illustrative Load Balancer console view: all backend instance groups reporting healthy across two zones.

Production Considerations

Cost: right-size node pools with autoscaling rather than static counts, use committed-use discounts for steady-state GKE and Cloud SQL capacity, and label every resource by environment and team for accurate cost attribution. Dev and staging rarely need the machine types or replica counts of prod—keep those variables distinct per environment rather than defaulting everything to production sizing out of caution.

High availability: run GKE as a regional cluster and Cloud SQL with REGIONAL availability in production; the global load balancer is already redundant across regions by design, so most of the HA work is in the two tiers behind it. A regional GKE cluster spreads its control plane and node pools across three zones automatically, so a single zone outage degrades capacity rather than taking the application down entirely.

Backup and DR: enable Cloud SQL point-in-time recovery and automated daily backups, and replicate Terraform state and critical application data to a second region on a schedule so a regional outage isn't also a data-loss event. Document and periodically rehearse the actual restore procedure—an untested backup is only a hypothesis about your recovery time, not a guarantee of it.

- **Security:** enforce private IP only for Cloud SQL, use Workload Identity for GKE Pod-to-GCP-API authentication instead of node service account keys, and give each module its own least-privilege IAM bindings—never share one broad service account across environments or components.

Taken together, these four concerns are why the folder structure and module boundaries matter so much: sizing, redundancy, and security posture are all just variables passed into the same modules, so tightening them for production never means rewriting the underlying resources. A team inheriting this project for the first time should be able to answer “what changes between environments” by reading `terraform.tfvars`, not by diffing hundreds of lines of duplicated resource definitions.

Key Takeaways

- A production system is a composition of small, well-tested modules—VPC, GKE, Cloud SQL, load balancer, storage, monitoring—not one monolithic configuration.
- The same environment-plus-shared-module pattern used throughout this book scales cleanly from a toy example to a real multi-tier application.
- Promote identical module versions from dev through staging to prod, validating with smoke tests and load tests at each stage before moving forward.
- Cost, high availability, backup/DR, and security are ongoing operational concerns—bake them into the modules and variables themselves, not into one-off console changes made after the fact.

Chapter 23: Terraform Interview Questions

Beginner Level Questions (1-16)

Q1. What is Terraform?

Terraform is an open-source Infrastructure as Code (IaC) tool created by HashiCorp that lets you define, provision, and manage cloud and on-premises infrastructure using a declarative configuration language called HCL (HashiCorp Configuration Language). Instead of clicking through a cloud console, you write `.tf` files describing the desired end state of resources such as VMs, networks, and databases, and Terraform figures out how to create, update, or delete real infrastructure to match that description. It is provider-agnostic, supporting GCP, AWS, Azure, Kubernetes, and hundreds of other platforms through plugins called providers.

Q2. How do you install Terraform and verify the installation?

On Linux, you typically download the binary from the HashiCorp releases page or install it via a package manager like `apt` using the HashiCorp APT repository, or via tools like `tfenv` for version management. After installation, running `terraform version` in a terminal confirms the installed version and lists any loaded provider versions. It's good practice to pin a specific Terraform version in your project using a `required_version` constraint inside the `terraform` block.

Q3. What does terraform apply do?

`terraform apply` executes the actions proposed in a Terraform plan to reach the desired state, actually creating, modifying, or destroying real infrastructure resources. By default it will show the plan and prompt for confirmation (yes) before proceeding, unless run with `-auto-approve`. It then updates the state file to reflect the newly applied infrastructure.

Q4. What is the google provider and how is it configured?

The `google` provider is the official HashiCorp/GCP-maintained plugin that exposes GCP resources and data sources to Terraform. A basic configuration looks like:

```
provider "google" {
  project = "my-gcp-project"
  region  = "us-central1"
  zone    = "us-central1-a"
}
```

There is also a `google-beta` provider that exposes newer or beta GCP features not yet promoted to the general availability provider.

Q5. What is terraform.tfvars?

`terraform.tfvars` is a special file Terraform automatically loads to assign values to declared input variables without needing to pass them on the command line. You can also use `*.auto.tfvars` files, which are loaded automatically, or explicitly point to a file with `-var-file`. This is useful for separating environment-specific values (like project IDs) from the reusable configuration logic.

Q6. Why does Terraform state matter?

State matters because without it, Terraform would have no way of knowing which real resources correspond to which configuration blocks, forcing it to either re-create everything or query every possible resource on every run. State enables efficient diffing, resource tracking, dependency management, and performance at scale. Because it can contain sensitive data (like passwords in plaintext), it must be stored and secured carefully, ideally in a remote, encrypted backend rather than a local file.

Q7. What is a data source in Terraform?

A data source lets Terraform read information about existing infrastructure that is not managed by the current Terraform configuration, such as looking up an existing VPC network or the latest OS image. It is declared with a data block, for example `data "google_compute_image" "debian" { family = "debian-12" project = "debian-cloud" }`, and referenced with `data.google_compute_image.debian.self_link`.

Q8. What are GCP regions and zones?

A region is a specific geographic location (like `us-central1`) that contains multiple isolated zones (like `us-central1-a`, `us-central1-b`), each representing an independent failure domain with its own power, cooling, and networking. Regional resources (like regional subnets) span all zones in a region for higher availability, while zonal resources (like a Compute Engine VM) exist in a single zone. Choosing regions close to users reduces latency, and spreading resources across zones improves resilience.

Q9. What does terraform validate do?

`terraform validate` checks whether a configuration is syntactically valid and internally consistent, such as verifying required arguments are present and that references between resources resolve correctly. It does not contact any cloud provider API or check whether credentials work, so it is fast and safe to run in CI as an early sanity check before `plan`.

Q10. How does Terraform infer dependencies between resources?

Terraform builds an internal dependency graph automatically whenever one resource's argument references another resource's attribute, such as a subnet's `network` argument referencing a `google_compute_network.self_link`. This implicit dependency ensures Terraform creates, updates, or destroys resources in the correct order. When there is no attribute reference but an ordering constraint still exists, you can use the explicit `depends_on` meta-argument.

Q11. What is the Terraform Registry?

The Terraform Registry is a public repository (registry.terraform.io) of providers and reusable modules published by HashiCorp, cloud vendors, and the community. It hosts the official `google` and `google-beta` providers as well as thousands of community modules, such as `terraform-google-modules/network/google`, which can be referenced directly in a module block by source and version.

Q12. What is `.terraform.lock.hcl`?

`.terraform.lock.hcl` is the dependency lock file that records the exact provider versions and cryptographic hashes selected during `terraform init`, ensuring consistent provider versions across team members and CI runs. Unlike `.terraform`, this file should be committed to version

control so that everyone uses identical provider versions until it is deliberately updated with `terraform init -upgrade`.

Q13. What is a firewall rule in GCP and how is it declared in Terraform?

A firewall rule (`google_compute_firewall`) controls the ingress or egress traffic allowed to and from VM instances within a VPC, based on criteria like source IP ranges, protocols, ports, and target tags or service accounts. Firewall rules in GCP are stateful and apply at the VPC level rather than per-subnet.

```
resource "google_compute_firewall" "allow_ssh" {
  name      = "allow-ssh"
  network   = google_compute_network.vpc.name
  allow {
    protocol = "tcp"
    ports    = ["22"]
  }
  source_ranges = ["0.0.0.0/0"]
}
```

Q14. What is the terraform block used for?

The `terraform` block is a top-level configuration block used to set Terraform-wide settings such as the required Terraform CLI version (`required_version`), the required providers and their version constraints (`required_providers`), and the backend configuration for storing state. It does not declare any infrastructure itself; it configures Terraform's own behavior.

```
terraform {
  required_version = ">= 1.5.0"
  required_providers {
    google = {
      source  = "hashicorp/google"
      version = "~> 5.0"
    }
  }
}
```

Q15. What is the purpose of the description argument in variables and outputs?

The `description` argument documents what a variable or output represents and is displayed when running `terraform plan` for undocumented required variables, in generated module documentation, and in tools like `terraform-docs`. It has no functional effect on execution but greatly improves the usability and maintainability of shared modules.

Q16. What is a Compute Engine instance and its basic Terraform resource type?

A Compute Engine instance is a virtual machine running in GCP, and it is declared with the `google_compute_instance` resource type. A minimal example specifies a `name`, `machine_type`, `zone`, `boot_disk` with a source image, and a `network_interface`.

```
resource "google_compute_instance" "vm" {
  name         = "web-vm"
  machine_type = "e2-medium"
```

```

zone          = "us-central1-a"
boot_disk {
  initialize_params {
    image = "debian-cloud/debian-12"
  }
}
network_interface {
  network = "default"
}
}

```

Intermediate Level Questions (1-16)

Q1. What is remote state and why use a GCS backend for it?

Remote state stores the terraform.tfstate file in a shared, durable location instead of on a single engineer's laptop, so that a whole team and CI/CD pipelines can safely collaborate on the same infrastructure. The GCS backend stores state as an object in a Cloud Storage bucket, supports state locking (via built-in mechanisms), versioning (via bucket object versioning), and can be encrypted with CMEK.

```

terraform {
  backend "gcs" {
    bucket = "my-tf-state-bucket"
    prefix = "envs/prod"
  }
}

```

Q2. How do variable validation blocks work?

A validation block inside a variable declaration lets you enforce custom constraints beyond basic type checking, using a condition expression and an error_message shown when the condition is false. This catches invalid input early, at plan time, rather than failing deep into a provider API call.

```

variable "environment" {
  type = string
  validation {
    condition     = contains(["dev", "staging", "prod"], var.environment)
    error_message = "environment must be dev, staging, or prod."
  }
}

```

Q3. Give an example contrasting count with for_each for the same use case.

Using count with a list, deleting the first element shifts all resource indices, causing Terraform to plan destroy/recreate for every subsequent resource:

```

resource "google_compute_instance" "vm" {
  count = length(var.vm_names)
  name  = var.vm_names[count.index]
}

```

Using `for_each` with a set avoids this because each resource is tracked by its name, not its position:

```
resource "google_compute_instance" "vm" {
  for_each = toset(var.vm_names)
  name     = each.value
}
```

Q4. When is `depends_on` needed if Terraform already infers dependencies automatically?

`depends_on` is needed when there is a dependency that isn't visible through attribute references, such as an IAM permission that must exist before an API call succeeds, or ordering requirements enforced by an external system rather than data flow. It's a blunt, explicit tool that should be used sparingly, since overuse makes the dependency graph harder to reason about and can slow down parallel applies.

```
resource "google_project_iam_member" "sa_role" {
  project = var.project_id
  role    = "roles/editor"
  member  = "serviceAccount:${google_service_account.app.email}"
}

resource "google_compute_instance" "vm" {
  depends_on = [google_project_iam_member.sa_role]
  ...
}
```

Q5. What does `prevent_destroy` protect against, and what are its limitations?

`prevent_destroy = true` causes Terraform to error out and refuse to destroy the resource, whether directly via `terraform destroy` or indirectly through a change that requires replacement. It's commonly used on critical resources like production databases. Its limitation is that it only protects at the Terraform-plan level; someone can still remove the block from configuration first, or delete the resource manually outside Terraform.

Q6. What is `terraform import` used for?

`terraform import` brings an existing resource that was created outside Terraform (e.g., manually in the console) under Terraform management, by associating it with a resource address in your state file. You must first write a matching resource block in configuration, then run something like `terraform import google_storage_bucket.assets my-bucket-name`, after which `plan` will show any drift between the imported resource's real configuration and your written HCL.

Q7. How do you provision a basic GKE cluster and node pool with Terraform?

You typically use `google_container_cluster` for the control plane and a separate `google_container_node_pool` for worker nodes, since managing node pools separately gives more control over scaling and upgrades. A common pattern removes the cluster's default node pool immediately since it's harder to customize.

```
resource "google_container_cluster" "primary" {
  name = "prod-cluster"
```

```

    location          = "us-central1"
    remove_default_node_pool = true
    initial_node_count  = 1
  }

  resource "google_container_node_pool" "primary_nodes" {
    name        = "primary-pool"
    cluster     = google_container_cluster.primary.name
    location    = "us-central1"
    node_count  = 3
    node_config {
      machine_type = "e2-standard-4"
    }
  }
}

```

Q8. What is the difference between additive and authoritative IAM resources in Terraform?

Additive resources like `google_project_iam_member` only add or remove the specific grant they manage, coexisting safely with IAM changes made by other tools or teams. Authoritative resources like `google_project_iam_binding` (per role) or `google_project_iam_policy` (for the entire policy) will overwrite anything not explicitly declared in that resource, meaning using them alongside manual console changes or other automation causes those external grants to be removed on the next apply. Best practice in shared environments is to use additive `_member` resources unless you're certain you fully own that policy.

Q9. What is configuration drift and how do you detect it?

Drift occurs when the real infrastructure state diverges from what's recorded in Terraform state, often due to manual console changes, another automation tool, or an external actor like autoscaling. You detect drift by running `terraform plan` periodically (or via scheduled CI jobs), which refreshes and compares live resource attributes against state, surfacing any unexpected diffs even when no configuration changes were made.

Q10. What is `count.index` and how is it used?

`count.index` is the zero-based index of the current instance when a resource or module uses the `count` meta-argument, letting you compute per-instance values like names or IPs.

```

resource "google_compute_instance" "vm" {
  count = 3
  name  = "web-${count.index}"
}

```

Q11. How do you pass outputs from one module as inputs to another module?

You reference the first module's declared output value directly as an argument to the second module's input variable, and Terraform automatically infers the dependency so the first module is applied before the second.

```

module "network" {
  source = "../modules/network"
}

```

```

}

module "gke" {
  source      = "../modules/gke"
  network_id = module.network.network_self_link
}

```

Q12. What is the purpose of terraform output -json in automation pipelines?

terraform output -json prints all declared outputs in structured JSON format, which downstream automation (like a deployment script or another CI job) can parse programmatically, for example to grab a newly created cluster's endpoint or a service account email for a subsequent step. This avoids fragile text-parsing of human-readable plan/apply output.

Q13. How do you reference a resource created with for_each from elsewhere in the configuration?

Because for_each produces a map of instances keyed by the iteration key, you reference a specific instance using resource_type.name["key"], not by numeric index like count resources.

```

resource "google_storage_bucket" "buckets" {
  for_each = toset(["logs", "assets"])
  name     = "${each.key}-bucket"
  location = "US"
}

output "logs_bucket_url" {
  value = google_storage_bucket.buckets["logs"].url
}

```

Q14. What is a terraform_remote_state data source used for?

It allows one Terraform configuration to read the outputs of another Terraform configuration's state file, enabling loosely coupled composition across separately-applied stacks, such as a networking stack's VPC ID being consumed by a separately managed application stack.

```

data "terraform_remote_state" "network" {
  backend = "gcs"
  config = {
    bucket = "my-tf-state-bucket"
    prefix = "envs/network"
  }
}

```

Q15. How do you enforce naming or tagging conventions across many resources without repeating code?

Common approaches include defining shared locals for standard labels/tags merged into every resource via the merge() function, or building a wrapper module that enforces the convention internally so consuming teams cannot bypass it.

```

locals {
  common_labels = {
    team      = var.team
    environment = var.environment
  }
}

resource "google_compute_instance" "vm" {
  labels = merge(local.common_labels, { role = "web" })
}

```

Q16. How can you safely rename a resource in configuration without destroying and recreating it?

Renaming a resource block's local name changes its address, and without further action, Terraform will plan to destroy the old address and create a new one. To preserve the underlying infrastructure, you either use `terraform state mv old_address new_address` before applying, or, in Terraform 1.1+, add a `moved` block declaring the old-to-new mapping so Terraform automatically treats it as a rename rather than a replace.

Advanced Level Questions (1-16)

Q1. What is Workload Identity Federation and why is it preferred for Terraform CI/CD over service account keys?

Workload Identity Federation lets an external identity provider (like GitHub Actions OIDC tokens) exchange short-lived tokens for GCP credentials without ever storing a long-lived service account JSON key, by configuring a workload identity pool and provider that trusts the external IdP's tokens and maps them to an impersonated service account. This eliminates the risk of leaked static keys, supports fine-grained conditions on which repository/branch can assume the identity, and aligns with zero standing-credential security practices.

Q2. How do you manage cross-project IAM and resources in a single Terraform run?

You configure multiple aliased `google` provider blocks, one per target project, and assign each resource the appropriate `provider = google.<alias>` (or set `project` explicitly on resources/data that support a `project` argument directly). Cross-project IAM grants — like allowing a service account in project A to access a bucket in project B — use the resource in project B's provider context while referencing the service account's fully-qualified email from project A, and typically require the caller's identity to have the necessary `resourceManager` or `serviceUsage` permissions across both projects.

Q3. How do you secure a Terraform state file stored in GCS, including CMEK encryption?

Beyond restricting bucket IAM to only the CI service account and a small set of admins, and enabling object versioning for recovery, you can configure the bucket with a Customer-Managed Encryption Key (CMEK) from Cloud KMS so that state data at rest is encrypted with a key you control and can rotate or revoke.

```

resource "google_storage_bucket" "tf_state" {
  name     = "my-tf-state"
  location = "US"
}

```

```
encryption {
  default_kms_key_name = google_kms_crypto_key.state_key.id
}
versioning {
  enabled = true
}
}
```

You should also ensure the KMS key's IAM grants the storage service agent roles/cloud-kms.cryptoKeyEncrypterDecrypter.

Q4. What is -parallelism and how does it affect large Terraform runs?

The -parallelism=n flag (default 10) controls how many resource operations Terraform executes concurrently during plan/apply. Increasing it can speed up applies with many independent resources, but can also trigger GCP API rate limits or quota errors if too many simultaneous API calls hit the same project; decreasing it can help work around such throttling at the cost of longer total runtime.

Q5. What is the run pipeline in Terraform Cloud / HCP Terraform?

A “run” progresses through stages: a VCS-triggered or CLI-triggered plan, followed by a policy check (Sentinel/OPA), a mandatory or optional human approval (a “confirm & apply” gate), then apply, all while Terraform Cloud manages remote state storage, locking, and variable/secret injection centrally. This provides consistent, auditable execution environments instead of relying on developer laptops or bespoke CI scripting for the actual Terraform binary execution.

Q6. What are key patterns for multi-region and multi-project architecture in Terraform on GCP?

Common patterns include a “hub-and-spoke” network topology using Shared VPC (host project with google_compute_shared_vpc_host_project and service projects attached via google_compute_shared_vpc_service_project), replicating stateless application stacks per region behind a global external HTTPS load balancer, and separating environments/business units into distinct GCP projects tied together through folder-level IAM and Shared VPC rather than one flat project. Terraform modules are typically parameterized by project ID and region so the same module is instantiated multiple times for each region/project combination.

Q7. What steps would you take to troubleshoot and recover from Terraform state corruption?

First, stop all further applies to prevent compounding the issue, and pull a known-good prior version of the state file from the GCS bucket's object versioning history. Use terraform state list/show to compare the restored state against reality, use terraform import to re-attach any resources that fell out of tracking, and validate with a careful terraform plan (not apply) to confirm no unintended destroy/recreate actions are proposed before resuming normal operations; as a last resort, resources can be manually reconciled via state rm plus import on a resource-by-resource basis.

Q8. How do you test Terraform code, and what is the difference between terraform test and Terratest?

`terraform test` (built into Terraform 1.6+) lets you write test files in HCL itself (`.tftest.hcl`) that run `plan` or `apply` against your configuration with specific variable values and assert on output/resource values, without leaving the Terraform tool ecosystem. Terratest is a Go testing library that actually applies real infrastructure in a temporary environment, runs validation logic (like HTTP requests against a deployed endpoint) against it, and then destroys it — giving true end-to-end confidence at the cost of real cloud spend and longer test run times, whereas `terraform test` can often run cheaper plan-only assertions.

```
run "valid_plan" {
  command = plan
  assert {
    condition      = google_compute_instance.vm.machine_type == "e2-medium"
    error_message = "unexpected machine type"
  }
}
```

Q9. Explain the failover mechanics of a highly available Cloud SQL instance managed by Terraform.

With `availability_type = "REGIONAL"`, GCP provisions a synchronous standby replica in a different zone of the same region; on primary zone failure, Cloud SQL automatically detects the outage and promotes the standby, updating the instance's connection endpoint transparently (typically within about 60 seconds, though this is managed by the GCP control plane rather than Terraform itself). Terraform's role here is provisioning the HA configuration correctly (`availability_type`, backup configuration, `deletion_protection`) — the failover event itself happens outside of any Terraform run, and afterward `terraform plan` should show no drift since the logical resource definition remains unchanged.

Q10. How would you manage hundreds or thousands of resources in Terraform without the plan becoming unmanageably slow or risky?

Strategies include splitting the configuration into multiple smaller state files by domain or environment (reducing per-plan graph size), using `-target` only for emergency scoped operations, tuning `-parallelism` to balance speed against API rate limits, and leveraging data-driven `for_each` over structured variables/YAML instead of hand-written repetitive resource blocks. Additionally, adopting a layered architecture (foundation/network stack, platform stack, application stack) means most day-to-day changes only touch the smallest, most frequently-changed layer rather than re-planning the entire estate each time.

Q11. How can Terraform be integrated with an approval-gated production deployment pipeline that also enforces policy checks?

A robust pipeline runs `terraform plan -out=plan.tfplan`, converts it with `terraform show -json plan.tfplan` for automated policy evaluation (OPA/Conftest or Sentinel), posts a human-readable summary for manual review/approval (e.g., a required reviewer gate in the CI system or Terraform Cloud's run UI), and only then executes `terraform apply plan.tfplan` using the exact reviewed plan file — ensuring policy checks, human approval, and the applied changes are all bound to the identical, unmodified plan artifact.

Q12. How do you handle a scenario where a `for_each` key needs to change (e.g., renaming a map key) without destroying the underlying resource?

Since `for_each` resources are tracked by key, changing a map's key is equivalent to removing the old key's instance and adding a new one, which would destroy and recreate the resource by default. To preserve the resource, use a `moved` block (or `terraform state mv`) mapping the old keyed address to the new keyed address, for example `moved { from = google_compute_instance.vm["old-key"] to = google_compute_instance.vm["new-key"] }`, before applying the renamed configuration.

Q13. What role does `terraform show -json` play in building custom tooling around Terraform?

`terraform show -json` (applied to either a state file or a saved plan) emits a fully machine-readable JSON representation of resources, their attributes, and planned changes, which is the foundation for building custom policy checks, cost estimation, drift dashboards, or compliance reporting tools without needing to parse HCL or human-readable CLI output directly.

Q14. What is a practical approach to enforcing “least privilege” IAM through Terraform at scale across many projects?

Rather than granting broad predefined roles project-wide, define custom roles scoped to exactly the permissions each workload identity needs, grant them at the narrowest resource scope possible (bucket-level or dataset-level rather than project-level) using additive `_member` resources, and centralize the definitions of these narrow roles in a shared, versioned module so consistency is maintained across dozens or hundreds of projects rather than each team inventing ad hoc roles.

Q15. What considerations go into deciding whether to use Terraform versus a GCP-native deployment tool (e.g., Deployment Manager successor tools, `gcloud`, or Config Connector)?

Terraform's multi-cloud support, mature ecosystem, and large community make it attractive for organizations with multi-cloud needs or existing Terraform expertise, while GCP-native or Kubernetes-native options like Config Connector integrate more tightly with GitOps/Kubernetes-centric workflows and can manage GCP resources as Kubernetes custom resources reconciled continuously (rather than only on-demand like Terraform apply). The choice often comes down to existing team skillset, whether infrastructure needs continuous reconciliation (favoring Config Connector's operator model) versus point-in-time apply workflows (favoring Terraform), and multi-cloud requirements.

Q16. How do `terraform plan` and policy-as-code tools interact with dynamically generated `for_each` resources in terms of auditability?

Because `for_each` resource instances are addressed by key rather than fixed line-by-line blocks, policy tools evaluating the JSON plan output need to iterate over each generated instance's planned attribute values individually (e.g., checking every bucket generated by a `for_each` for public access settings), rather than assuming a fixed, enumerable list of resource blocks matching the source code line count — this is naturally handled by JSON-based policy engines like OPA/Conftest since the plan JSON always fully enumerates every concrete instance regardless of how it was generated.

Chapter 24: Terraform Commands Cheat Sheet

This chapter is a fast, practical reference for the Terraform CLI commands you will use every day when working with Google Cloud Platform. Rather than re-explaining concepts already covered earlier in the book, each section below groups related commands into a table with the command syntax, its purpose, and a realistic GCP example. Keep this chapter bookmarked — it is designed to be scanned, not read cover to cover.

TIP

Run `terraform -help` or `terraform <subcommand> -help` at any time to see the full, version-specific flag list for your installed CLI. Flags do change slightly between major Terraform releases.

Project Setup Commands

These commands initialize a working directory, download providers and modules, and configure the backend that stores your state file.

Command	Purpose	Example
<code>terraform init</code>	Initializes a working directory: downloads the required providers (e.g., hashicorp/google), installs referenced modules, and sets up the backend. Must be run before <code>plan</code> or <code>apply</code> .	<code>terraform init</code>
<code>terraform init -upgrade</code>	Re-initializes and upgrades all provider and module versions to the newest ones allowed by your version constraints, updating <code>.terraform.lock.hcl</code> .	<code>terraform init -upgrade</code>
<code>terraform init -migrate-state</code>	Re-initializes with a new or changed backend configuration and automatically copies existing state into the new backend.	<code>terraform init -migrate-state</code> (used when moving from local state to a GCS backend)
<code>terraform init -reconfigure</code>	Re-initializes the backend from scratch, ignoring any existing saved backend configuration, without attempting to migrate state.	<code>terraform init -reconfigure</code> (used when the backend bucket was recreated and old state should be ignored)

NOTE

For GCP projects, `terraform init` is also the moment Terraform authenticates against the `google` and `google-beta` providers using your configured credentials (service account key, Application Default Credentials, or Workload Identity Federation).

Validation & Formatting

Use these commands to keep your .tf files syntactically valid and consistently formatted before every commit.

Command	Purpose	Example
terraform fmt	Rewrites .tf and .tfvars files in the current directory to the canonical Terraform style (indentation, alignment, spacing).	terraform fmt
terraform fmt -recursive	Applies formatting to the current directory and all subdirectories, useful in multi-module repositories.	terraform fmt -recursive
terraform validate	Checks that configuration files are syntactically valid and internally consistent (correct argument names, types, references), without contacting GCP.	terraform validate

BEST PRACTICE

Wire `terraform fmt -recursive -check` and `terraform validate` into a pre-commit hook or CI pipeline so malformed configuration never reaches a shared branch.

Planning & Applying

These are the core commands you will run constantly: previewing changes and then applying them to your GCP project.

Command	Purpose	Example
terraform plan	Computes and displays the execution plan — what Terraform would create, change, or destroy — without touching real infrastructure.	terraform plan
terraform plan -out=tfplan	Saves the computed plan to a binary file so it can be reviewed and later applied exactly as approved.	terraform plan -out=tfplan
terraform apply	Generates a fresh plan and prompts for confirmation before creating, updating, or destroying resources.	terraform apply
terraform apply tfplan	Applies a previously saved plan file exactly as it was computed, with no re-evaluation or confirmation prompt.	terraform apply tfplan

Command	Purpose	Example
<code>terraform apply -auto-approve</code>	Applies changes without an interactive yes/no prompt — intended for CI/CD pipelines, not interactive use.	<code>terraform apply -auto-approve</code>
<code>terraform apply -var="..."</code>	Passes a single input variable value on the command line, overriding any default.	<code>terraform apply -var="project_id=my-gcp-project-123456"</code>
<code>terraform apply -target=<resource></code>	Restricts the apply to a specific resource (and its dependencies), useful for troubleshooting or staged rollouts.	<code>terraform apply -target=google_compute_instance.web_s</code>
<code>terraform apply -replace=<resource></code>	Forces Terraform to destroy and recreate a specific resource, even if its configuration has not changed.	<code>terraform apply -replace=google_compute_instance.web_s</code>

WARNING

`-auto-approve` skips the safety confirmation step. Never use it against production state unless the pipeline has its own independent review gate (a merge-request approval, a manual pipeline trigger, or a plan-then-approve stage).

WARNING

`-target` is a scalpel, not a habit. Repeated use of `-target` in normal workflows is a sign that your configuration should be broken into smaller, better-isolated modules or state files.

Destroying Resources

Use these commands with care — they permanently remove infrastructure.

Command	Purpose	Example
<code>terraform destroy</code>	Destroys every resource tracked in the current state file, after showing a plan and asking for confirmation.	<code>terraform destroy</code>
<code>terraform destroy -target=<resource></code>	Destroys only the specified resource (and anything that depends on it), leaving the rest of the state intact.	<code>terraform destroy -target=google_sql_database_instance.</code>

WARNING

`terraform destroy` deletes real, billed GCP resources including data disks, Cloud SQL instances, and GKE clusters. Always run `terraform plan -destroy` first in shared environments, and confirm you are pointed at the intended workspace and backend before confirming.

State Management Commands

These commands inspect and directly manipulate the Terraform state file — the source of truth mapping your configuration to real GCP resource IDs.

Command	Purpose	Example
<code>terraform state list</code>	Lists every resource address currently tracked in state.	<code>terraform state list</code>
<code>terraform state show <resource></code>	Prints the full attributes Terraform has recorded for a single resource.	<code>terraform state show google_compute_instance.web_server</code>
<code>terraform state mv <src> <dst></code>	Renames a resource address in state, or moves it into/out of a module, without destroying and recreating the underlying infrastructure.	<code>terraform state mv google_compute_instance.web google_compute_instance.web_server</code>
<code>terraform state rm <resource></code>	Removes a resource from state without destroying it in GCP — Terraform simply forgets about it.	<code>terraform state rm google_storage_bucket.legacy_bucket</code>
<code>terraform state pull</code>	Downloads and prints the current remote state as raw JSON, useful for scripting or manual inspection.	<code>terraform state pull > state-backup.json</code>
<code>terraform state push</code>	Uploads a local state file to overwrite the remote state — a recovery operation, used sparingly.	<code>terraform state push state-backup.json</code>
<code>terraform force-unlock <LOCK_ID></code>	Manually removes a stuck state lock, typically after an interrupted apply left the GCS backend lock in place.	<code>terraform force-unlock 1b2c3d4e-5f6a-7b8c-9d0e-1f2a3b4c5d6e</code>

BEST PRACTICE

Always run `terraform state pull` to capture a backup before any `state mv`, `state rm`, or `state push` operation. These commands edit state directly and mistakes can be hard to reverse.

Workspace Commands

Workspaces let a single configuration manage multiple, isolated state files — commonly used for dev, staging, and prod environments on GCP.

Command	Purpose	Example
<code>terraform workspace list</code>	Lists all workspaces in the current backend, marking the active one with an asterisk.	<code>terraform workspace list</code>
<code>terraform workspace new <name></code>	Creates a new workspace with its own isolated state.	<code>terraform workspace new staging</code>

Command	Purpose	Example
<code>terraform workspace select <name></code>	Switches the active workspace.	<code>terraform workspace select prod</code>
<code>terraform workspace show</code>	Prints the name of the currently selected workspace.	<code>terraform workspace show</code>
<code>terraform workspace delete <name></code>	Deletes a workspace, which must have empty state first.	<code>terraform workspace delete staging</code>

NOTE

Workspaces change which state file is used but do not change which `.tf` files are read. Combine `terraform.workspace` with conditionals or `.tfvars` files per environment, as covered in Chapter 12.

Inspecting & Debugging

Use these commands to understand what Terraform currently knows, and to troubleshoot when something goes wrong.

Command	Purpose	Example
<code>terraform show</code>	Displays the resources and attributes currently recorded in state, in human-readable form.	<code>terraform show</code>
<code>terraform output</code>	Prints the values of declared output variables, such as a load balancer IP or a GKE cluster endpoint.	<code>terraform output cluster_endpoint</code>
<code>terraform graph</code>	Generates a DOT-format dependency graph of resources, which can be piped into Graphviz for visualization.	<code>terraform graph \ dot -Tpng > graph.png</code>
<code>terraform providers</code>	Lists the providers required by the configuration and where each requirement comes from.	<code>terraform providers</code>
<code>terraform providers schema -json</code>	Dumps the full machine-readable schema (every argument and attribute) for every provider in use.	<code>terraform providers schema -json > schema.json</code>
<code>TF_LOG=DEBUG terraform apply</code>	Enables verbose debug logging, including raw API calls to GCP, for diagnosing provider or authentication issues.	<code>TF_LOG=DEBUG terraform apply 2> debug.log</code>

TIP

Set `TF_LOG_PATH=./terraform.log` alongside `TF_LOG=DEBUG` to write logs to a file instead of flooding your terminal, which is especially useful when debugging a long GKE cluster creation.

Import & Migration

Bring existing, manually-created GCP resources under Terraform management using either the classic `import` command or the modern `import` block.

Command	Purpose	Example
<code>terraform import <address> <id></code>	Imports a single existing resource into state at the given resource address, matched to its GCP resource ID. Requires a matching resource block to already exist in configuration.	<pre>terraform import google_compute_instance.web_server projects/my-gcp-project- 123456/zones/us-central1- a/instances/web-server-1</pre>

The newer declarative `import` block (Terraform 1.5+) lets you plan an import before it happens, and keep the mapping as version-controlled code:

```
import {
  to = google_compute_instance.web_server
  id =
↪ "projects/my-gcp-project-123456/zones/us-central1-a/instances/web-server-1"
}

resource "google_compute_instance" "web_server" {
  name          = "web-server-1"
  machine_type  = "e2-medium"
  zone          = "us-central1-a"
  # remaining arguments generated via `terraform plan
↪ -generate-config-out=generated.tf`
}
```

TIP

Pair an `import` block with `terraform plan -generate-config-out=generated.tf` to have Terraform write a first draft of the resource configuration for you, rather than transcribing every argument by hand.

Module & Registry Commands

These commands manage locally cached copies of modules referenced by your configuration, including those pulled from the Terraform Registry or a private GCS-based module source.

Command	Purpose	Example
<code>terraform get</code>	Downloads and installs modules referenced in configuration that are not already present in <code>.terraform/modules</code> .	<code>terraform get</code>
<code>terraform get -update</code>	Re-downloads all referenced modules, updating to the latest matching version even if a cached copy already exists.	<code>terraform get -update</code>

NOTE

`terraform init` calls `terraform get` internally, so most day-to-day workflows never need to run `terraform get` directly. It is mainly useful when module sources changed but you do not want to re-run a full `init`.

Chapter 25: Summary

You began this book by asking a simple question: how do you manage cloud infrastructure without clicking through a console, keeping tribal knowledge in someone’s head, or hoping a shell script written two years ago still works? Twenty-two chapters and a full command reference later, you now have a real answer, backed by hands-on practice against actual Google Cloud projects. This closing chapter steps back from syntax and looks at the shape of the journey as a whole, then points forward to how you can keep building on it.

Key Takeaways

The book opened with the case for Infrastructure as Code itself: the shift from manually provisioned, drift-prone environments to declarative configuration that can be reviewed, versioned, and reproduced on demand. You saw why Terraform in particular has become the dominant multi-cloud tool for this job, and how its declarative model differs fundamentally from imperative scripting — you describe the desired end state, and Terraform figures out the sequence of API calls needed to get there. From that foundation, the early chapters walked through setting up a Google Cloud account, enabling billing and APIs, installing the Terraform CLI, and authenticating the `google` provider with service accounts and Application Default Credentials, so that every example afterward had solid ground to stand on.

With the environment ready, the middle chapters built up the core mechanics that everything else depends on: resources, data sources, providers, the dependency graph, and the plan-apply lifecycle that makes Terraform safe to run against production. You then went a layer deeper into state — what it is, why it must be stored remotely and locked during team use, and how a Google Cloud Storage backend with object versioning gives you both durability and a safety net. Variables, locals, and outputs gave you the vocabulary to make configuration reusable rather than hard-coded, and modules turned that vocabulary into genuinely shareable, composable building blocks, whether pulled from the public Terraform Registry or authored in-house for your organization’s own conventions.

From there, the book turned to the resources that make up a real GCP footprint. You provisioned Compute Engine instances and instance templates, built networking with VPCs, subnets, and firewall rules, stood up Google Kubernetes Engine clusters and node pools, managed Cloud Storage buckets and lifecycle rules, and configured Cloud SQL instances with private connectivity and automated backups. Layered on top of all of that was Identity and Access Management: service accounts, custom roles, and the principle of least privilege applied consistently across every resource type, so that “who can do what” became part of the same version-controlled configuration as the infrastructure itself.

Workspaces and environment strategy addressed a question every team eventually faces: how do you run the same configuration safely across dev, staging, and production without duplicating code or risking a staging mistake reaching production? You paired that with CI/CD pipelines — running `terraform plan` on every pull request, gating `apply` behind human approval, and treating your pipeline itself as the enforcement mechanism for good practice rather than relying on discipline alone. Security wove through this entire section: secret management with tools like Secret Manager rather than plaintext variables, remote state encryption, and scanning configuration for misconfigurations before they ever reach a live project.

The more advanced chapters gave you the language features that separate competent Terraform from expert Terraform: `for_each` and `count` for dynamic resource creation, complex type constraints and validation blocks, dynamic blocks for repeating nested configuration, and functions for transforming data inline. These are the tools that let a configuration scale from ten resources to hundreds without becoming unreadable. The book’s capstone chapter then pulled every one of these threads together into a single enterprise-grade, production-style project — multiple environments, remote state, modules, CI/CD, IAM, and security controls all operating as one coherent system, mirroring what you would actually be asked to build and defend in a real organization.

Taken together, these chapters were never just about learning a syntax. They were about learning to think in terms of desired state, reproducibility, and reviewable change — a mindset that will keep paying off even as the specific tools around it evolve.

Learning Roadmap

If you are working through this material from a standing start, here is a realistic stage-by-stage progression to take you from zero to production-ready over roughly 8 to 12 weeks. Adjust the pace to your own availability, but resist the urge to skip the hands-on exercises in the earlier chapters — they are what the later, harder chapters depend on.

- **Weeks 1-2: Fundamentals and first project.** Read the IaC and Terraform overview chapters, install the Terraform CLI and the Google Cloud SDK, create a GCP project with billing enabled, and get comfortable with `init`, `plan`, and `apply` by deploying a single Compute Engine instance end to end.
- **Weeks 3-4: State, variables, and modules.** Move from local state to a GCS backend with locking, practice variables/locals/outputs until they feel natural, and build your first reusable module (a simple VPC or a compute module is a good first candidate).
- **Weeks 5-6: Core GCP services.** Provision networking (VPCs, subnets, firewall rules), stand up a small GKE cluster, work with Cloud Storage lifecycle rules, and deploy a Cloud SQL instance with private IP connectivity — the four services you will touch most often in real work.
- **Weeks 7-8: IAM, security, and CI/CD.** Design custom IAM roles and service accounts around least privilege, integrate Secret Manager for sensitive values, and wire up a CI/CD pipeline that runs `plan` on every pull request and gates `apply` behind approval.
- **Weeks 9-10: Advanced language features and workspaces.** Practice `for_each`, `count`, dynamic blocks, and validation rules until you can express repetitive infrastructure concisely, and set up workspaces to manage dev, staging, and production from a single configuration.
- **Weeks 11-12: Capstone project and interview preparation.** Rebuild the book's enterprise capstone project from memory rather than by copying it, extend it with one feature of your own choosing, and use the command cheat sheet in Chapter 24 as a rapid-fire self-quiz before interviews or certification exams.

TIP

At every stage, destroy what you build once you are done experimenting. GCP resources like Cloud SQL instances and GKE clusters continue to incur cost while they exist, running `terraform destroy` at the end of a study session is both good practice and good budget management.

Recommended Certifications

Certifications are not a substitute for hands-on experience, but they are an efficient way to validate your knowledge, structure your remaining study, and signal competence to employers. The following three, taken together, map closely onto everything covered in this book.

Google Associate Cloud Engineer

This is Google Cloud's foundational, hands-on certification, and it is the natural first stop for anyone who has worked through this book's early GCP chapters. It validates the ability to deploy applications, monitor operations, and manage enterprise solutions on Google Cloud — covering Compute Engine, Kubernetes Engine, Cloud Storage, IAM, and networking at a practical, day-to-day operator level. There are no formal prerequisites, though Google recommends around six months of hands-on GCP experience, which the exercises in Chapters 1 through 12 of this book go a long way toward providing. For a Terraform practitioner, this certification proves you understand the resources you are provisioning, not just the HCL syntax used to provision them — a distinction interviewers care about.

Google Professional Cloud DevOps Engineer

This is a professional-level certification and noticeably more demanding than the Associate exam, generally recommended only after some real production experience. It validates the ability to implement CI/CD pipelines, manage services using SRE principles, and apply IaC practices at scale — directly overlapping with this book’s chapters on workspaces, CI/CD, and the enterprise capstone project. Google recommends around three or more years of industry experience, including one or more years working specifically with GCP, before attempting it. For someone who has built out the workspace strategy and CI/CD pipeline sections of this book, this certification is the natural next milestone: it rewards exactly the operational, pipeline-and-process thinking those chapters were designed to teach.

HashiCorp Certified: Terraform Associate

This is the tool-specific counterpart to the two GCP certifications above, and the most directly aligned with this book’s content overall. It validates practical knowledge of Terraform itself: the CLI workflow, state management, modules, provisioners, workspaces, and the underlying configuration language, independent of any single cloud provider. It has no formal prerequisites and is considered an entry- to intermediate-level exam, making it approachable relatively early, though it will feel far more natural after completing the state, variables/modules, and advanced language feature chapters of this book. Because the exam is cloud-agnostic, it also signals that your Terraform skills transfer beyond GCP — a valuable credential if your career takes you toward multi-cloud or platform engineering roles.

Together, these three certifications form a natural progression: the Associate Cloud Engineer exam proves you understand the cloud platform itself, the Professional Cloud DevOps Engineer exam proves you can operate and automate that platform at scale and under production constraints, and the Terraform Associate exam proves deep, transferable mastery of the tool you used to do it. Pursued in that order, each one reinforces the material from the certification before it, and together they map almost exactly onto the arc of this book.

Thank you for working through this guide from Chapter 1 to the very end. Terraform and Google Cloud are both large, fast-moving ecosystems, and no single book can cover every service or every edge case you will encounter in production. What this book has tried to give you instead is the mental model, the habits, and the hands-on reps to keep learning confidently on your own. Keep building, keep destroying what you no longer need, and keep reading the plan output before you type “yes.” Good luck out there.